System I

Foundations of Digital Logic

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Overview

- Introduction
- Binary logic and gates
- Transistors
- Some IC parameters
- Boolean algebra
- Logic functions
- Simplification of logic functions
- Additional Gates and Circuits

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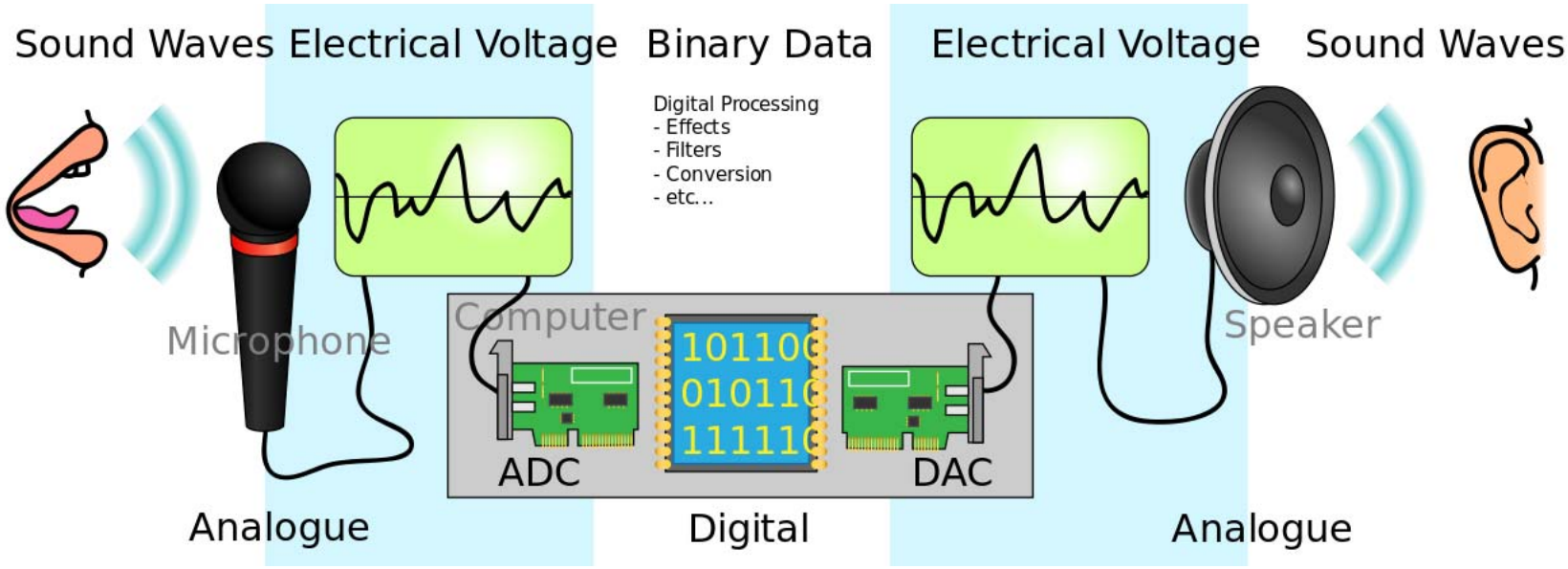
Electronic Systems



Analog signals vs. Digital signals

Why Digital?

- Intentionally restrict design choices
- Example: Digital discipline
  - Discrete voltages instead of continuous
  - Simpler to design than analog circuits – can build more sophisticated systems
  - Digital systems replacing analog predecessors:
    - i.e., digital cameras, digital television, cell phones, CDs

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Binary Logic and Gates

- Binary variables take on one of two values.
- Logical operators operate on binary values and binary variables.
- Basic logical operators are the logic functions AND, OR and NOT.
- Logic gates implement logic functions.
- Boolean Algebra: a useful mathematical system for specifying and transforming logic functions.
- We study Boolean algebra as a foundation for designing and analyzing digital systems!

Binary Variables

- Recall that the two binary values have different names:
  - True/False
  - On/Off
  - Yes/No
  - 1/0
- We use 1 and 0 to denote the two values.
- Variable identifier examples:
  - A, B, y, z, or X

Logical Operations

- The three basic logical operations are:
  - AND
  - OR
  - NOT
- AND is denoted by a dot (·).
- OR is denoted by a plus (+).
- NOT is denoted by an overbar ($\overline{\ }$), a single quote mark (') after, or (~) before the variable.

Truth Table

- A tabular listing of the values of a function for all possible combinations of values on its arguments

- Example: Truth tables for the basic logic operations:

AND		
X	Y	$Z = X \cdot Y$
0	0	0
0	1	0
1	0	0
1	1	1

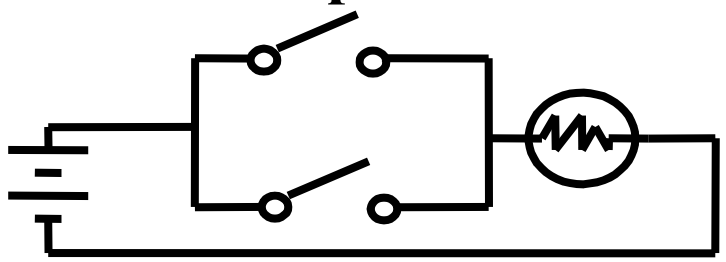
OR		
X	Y	$Z = X+Y$
0	0	0
0	1	1
1	0	1
1	1	1

NOT	
X	$Z = \overline{X}$
0	1
1	0

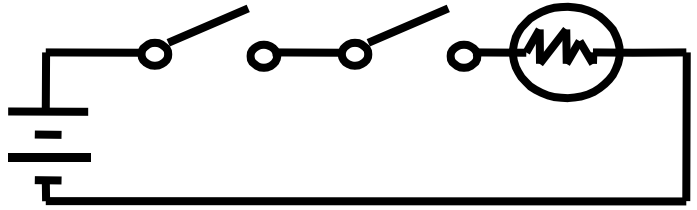
How to Model Logic Functions?

- Using Switches
  - For inputs:
    - logic 1 is switch closed
    - logic 0 is switch open
  - For outputs:
    - logic 1 is light on
    - logic 0 is light off.
  - NOT uses a switch such that:
    - logic 1 is switch open
    - logic 0 is switch closed

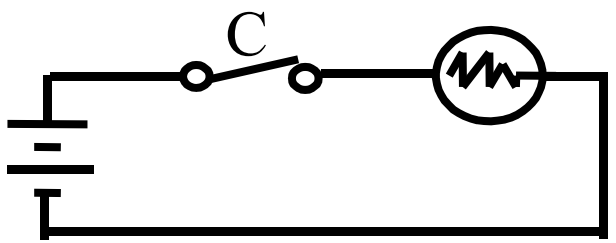
Switches in parallel => OR



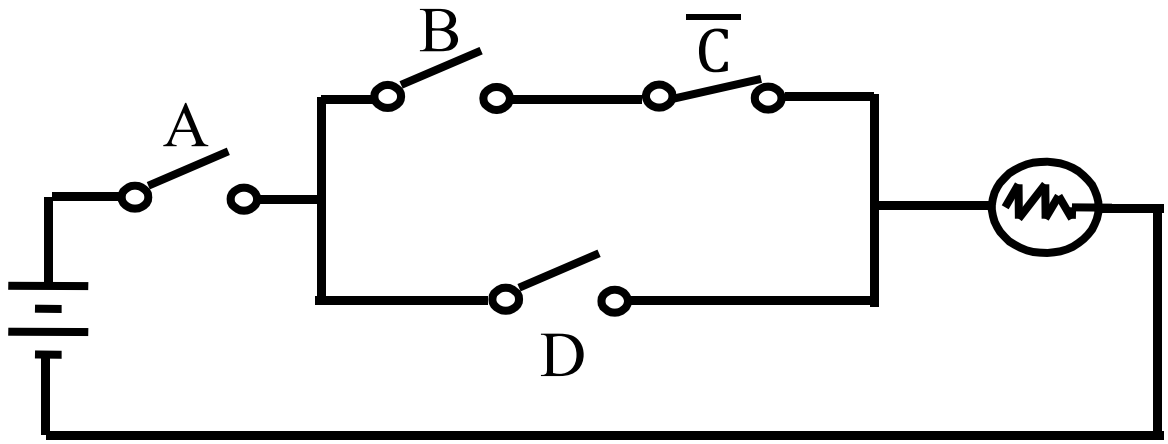
Switches in series => AND



Normally-closed switch => NOT



An Example: Logic Using Switches



- Light is on (L = 1) for

  L(A, B, C, D) =

  and off (L = 0), otherwise.
- Useful model for relay circuits and for CMOS gate circuits, the foundation of current digital logic technology

Logic Gates

- Perform logic functions:
  - inversion (NOT), AND, OR, NAND, NOR, etc.
- Single-input:
  - NOT gate, buffer
- Two-input:
  - AND, OR, XOR, NAND, NOR, XNOR
- Multiple-input

Other Gate Types

1. Graphics symbol and truth table for primitive gates

- Primitive gate is simple and fast

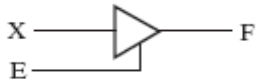
Name	Distinctive shape	Algebraic equation	Truth table
AND		F = XY	X Y | F 0 0 | 0 0 1 | 0 1 0 | 0 1 1 | 1
OR		F = X + Y	X Y | F 0 0 | 0 0 1 | 1 1 0 | 1 1 1 | 1
NOT (inverter)		F = $\overline{X}$	X | F 0 | 1 1 | 0
Buffer		F = X	X | F 0 | 0 1 | 1
3-State Buffer			E X | F 0 0 | Hi-Z 0 1 | Hi-Z 1 0 | 0 1 1 | 1
NAND		F = $\overline{X \cdot Y}$	X Y | F 0 0 | 1 0 1 | 1 1 0 | 1 1 1 | 0
NOR		F = $\overline{X + Y}$	X Y | F 0 0 | 1 0 1 | 0 1 0 | 0 1 1 | 0

Figure 2-28: primitive Digital Logic Gates

Other Gate Types

2. Graphics symbols and truth table for complex gates

- Complex circuit Lower the cost and transmit time in circuit

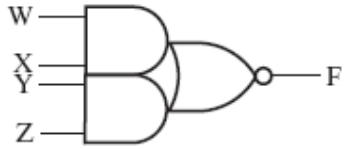
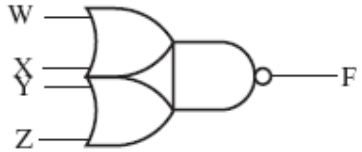
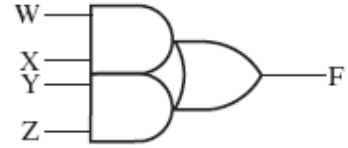
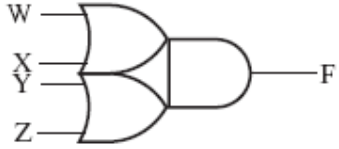
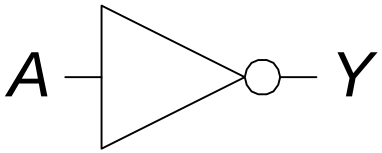
Name	Distinctive shape symbol	Algebraic equation	Truth table
Exclusive–OR (XOR)		$F = X\overline{Y} + \overline{X}Y$ $= X \oplus Y$	X Y | F 0 0 | 0 0 1 | 1 1 0 | 1 1 1 | 0
Exclusive–NOR (XNOR)		$F = XY + \overline{X}\overline{Y}$ $= \overline{X \oplus Y}$	X Y | F 0 0 | 1 0 1 | 0 1 0 | 0 1 1 | 1
AND-OR-INVERT (AOI)		$F = \overline{WX + YZ}$	
OR-AND -INVERT (OAI)		$F = \overline{(W + X)(Y + Z)}$	
AND-OR (AO)		$F = WX + YZ$	
OR-AND (OA)		$F = (W + X)(Y + Z)$	

Figure 2-29: Complex Digital Logic Gates

Single-Input Logic Gates

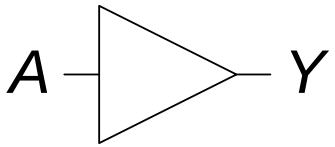
NOT



$Y = \overline{A}$

A	Y
0	
1	

BUF

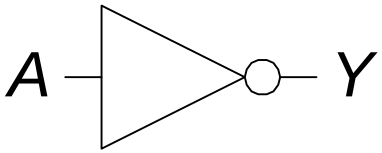


$Y = A$

A	Y
0	
1	

Single-Input Logic Gates

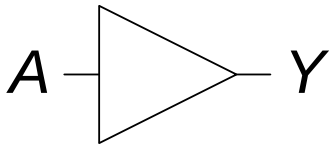
NOT



$Y = \overline{A}$

A	Y
0	1
1	0

BUF

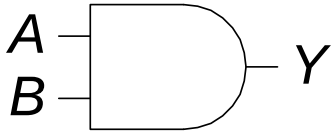


$Y = A$

A	Y
0	0
1	1

Two-Input Logic Gates

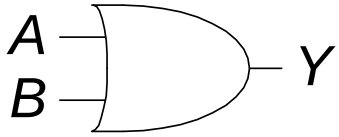
AND



$Y = AB$

A	B	Y
0	0	
0	1	
1	0	
1	1	

OR

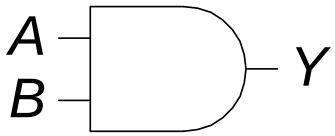


$Y = A + B$

A	B	Y
0	0	
0	1	
1	0	
1	1	

Two-Input Logic Gates

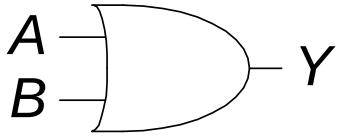
AND



$Y = AB$

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

OR

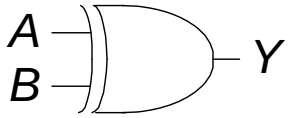


$Y = A + B$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

More Two-Input Logic Gates

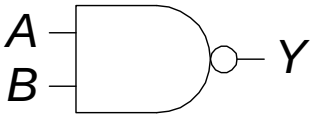
XOR



$Y = A \oplus B$

A	B	Y
0	0	
0	1	
1	0	
1	1	

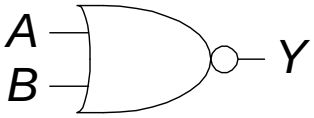
NAND



$Y = \overline{AB}$

A	B	Y
0	0	
0	1	
1	0	
1	1	

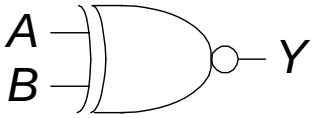
NOR



$Y = \overline{A + B}$

A	B	Y
0	0	
0	1	
1	0	
1	1	

XNOR

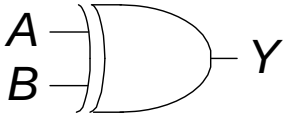


$Y = \overline{A \oplus B}$

A	B	Y
0	0	
0	1	
1	0	
1	1	

More Two-Input Logic Gates

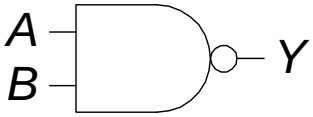
XOR



$Y = A \oplus B$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

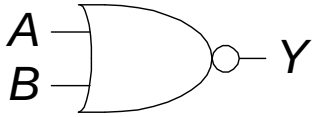
NAND



$Y = \overline{AB}$

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

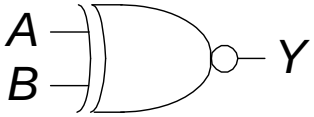
NOR



$Y = \overline{A + B}$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

XNOR

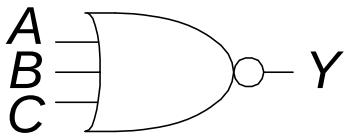


$Y = \overline{A \oplus B}$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Multiple-Input Logic Gates

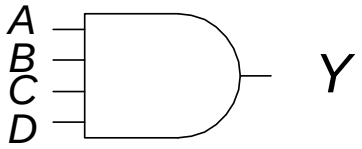
NOR3



$Y = \overline{A+B+C}$

A	B	C	Y
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

AND4

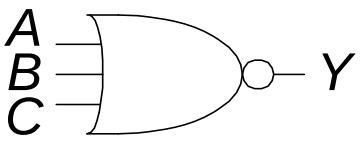


$Y = ABCD$

A	B	C	Y
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

Multiple-Input Logic Gates

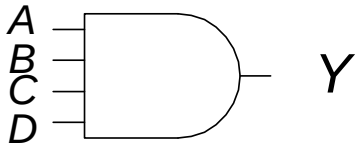
NOR3



$Y = \overline{A+B+C}$

A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

AND4



$Y = ABCD$

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

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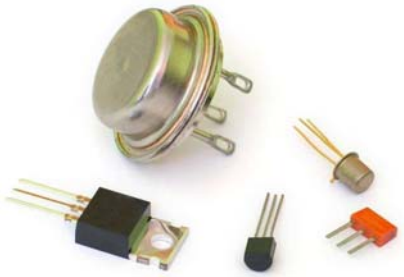
Relays vs. Vacuum Cube vs. Transistor

- In the earliest computers, switches were opened and closed by magnetic fields produced by energizing coils in *relays*. The switches in turn opened and closed the current paths.
- Later, *vacuum tubes* that open and close current paths electronically replaced relays.
- Today, *transistors* are used as electronic switches that open and close current paths.

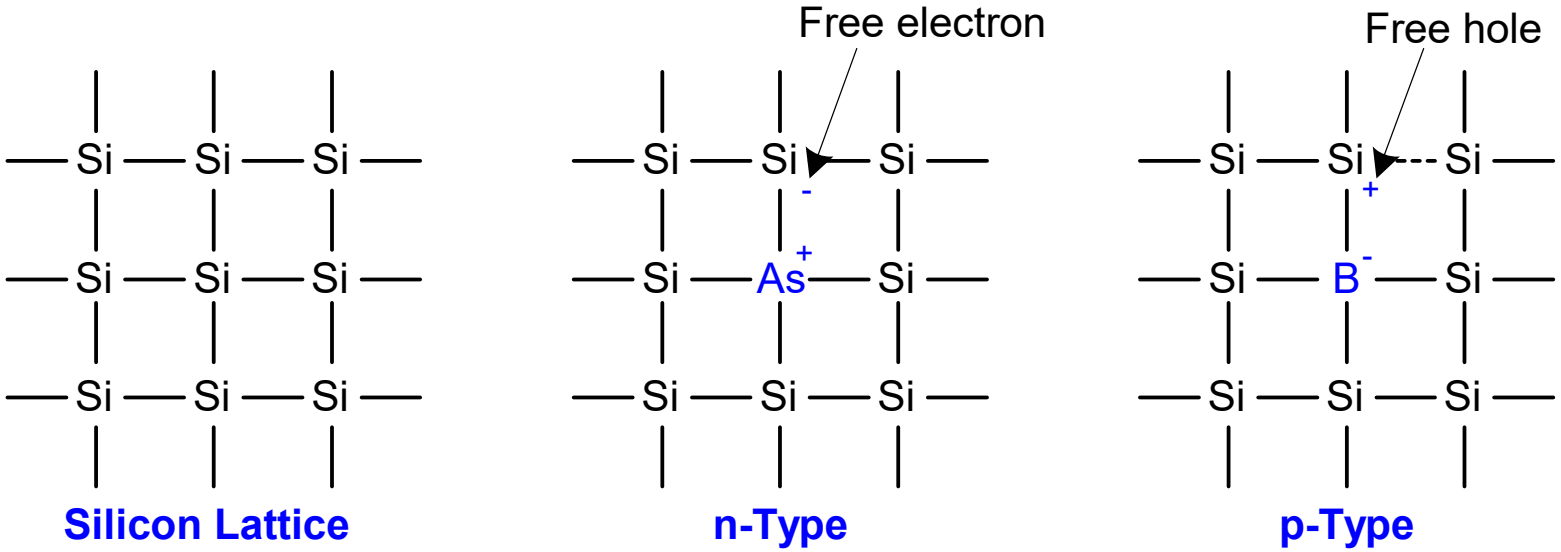


Robert Noyce, 1927-1990

- Nicknamed “Mayor of Silicon Valley”
- Cofounded Fairchild Semiconductor in 1957
- Cofounded Intel in 1968
- Co-invented the integrated circuit

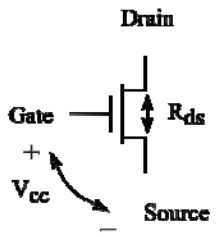
Silicon

- Transistors built from silicon, a semiconductor
- Pure silicon is a poor conductor (no free charges)
- Doped silicon is a good conductor (free charges)
  - n-type (free negative charges, electrons)
  - p-type (free positive charges, holes)

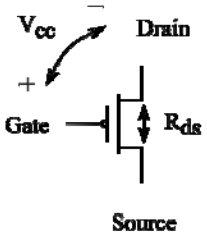


Integrated Circuit and Transistors

- Integrated Circuit (IC)
  - Transistor-Transistor Logic (TTL)
    - Bipolar Junction Transistor (BJT)
  - Complementary Metal-Oxide Semiconductor (CMOS)
    - Field Effect Transistor (FET)
- CMOS Transistors
  - Gate, source, drain
  - NMOS transistor / PMOS transistor

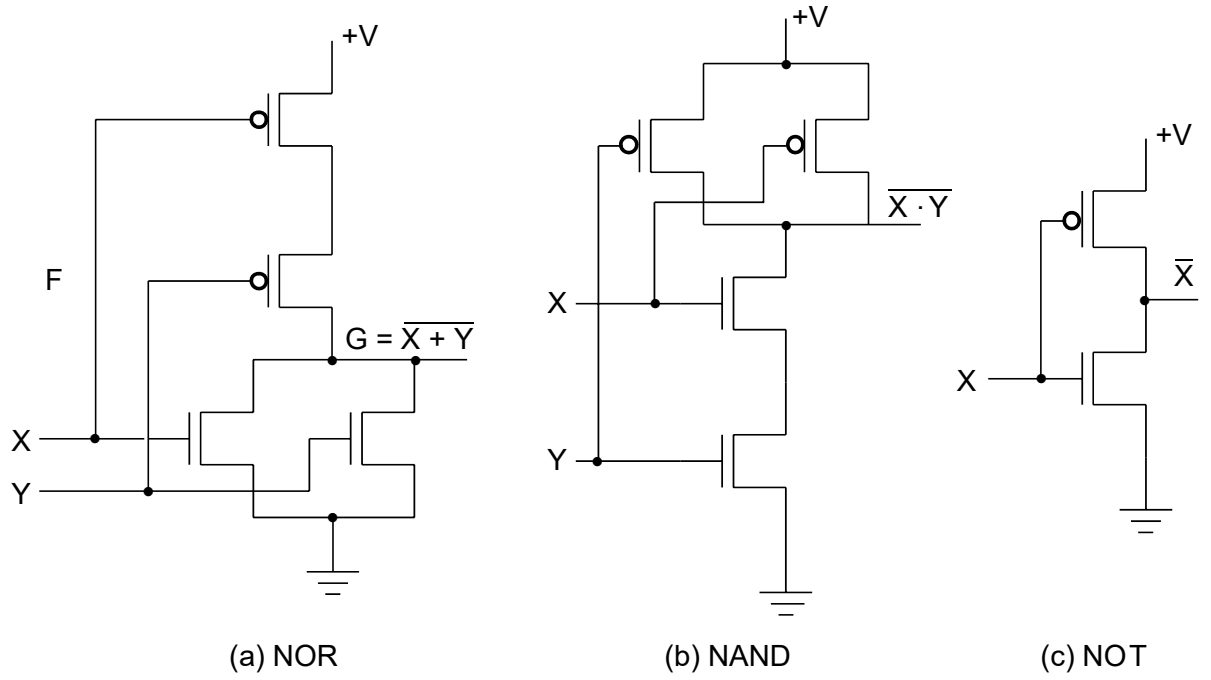


NMOS



PMOS

Implementation of Logic Gates with Transistors



- Transistor or tube implementations of logic functions are called logic gates or just gates
- Transistor gate circuits can be modeled by switch circuits

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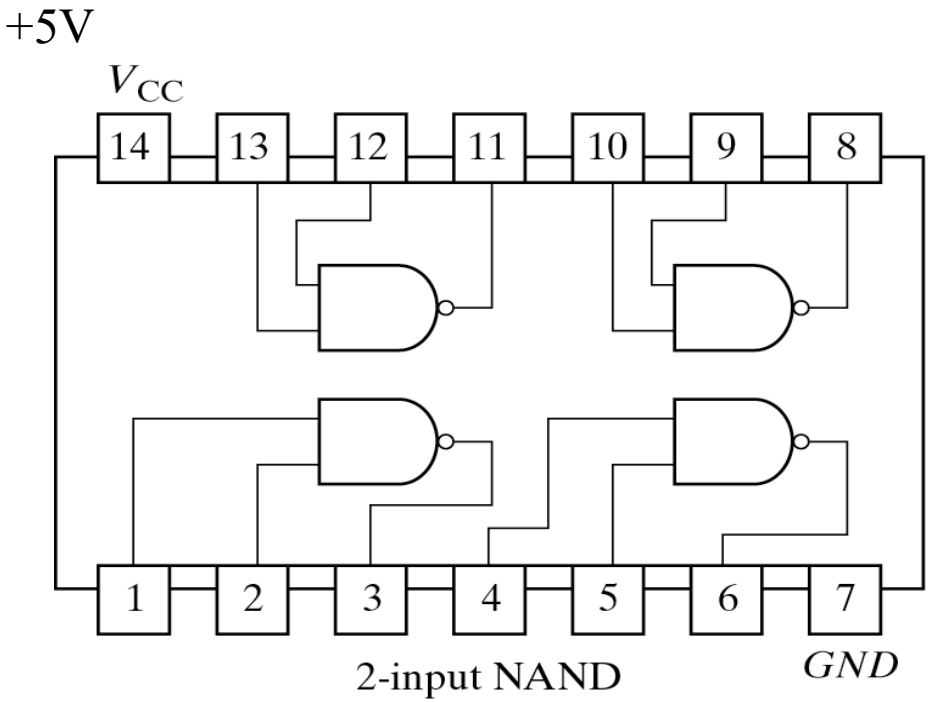
Some IC Parameters

- V_{CC} (Power Supply Voltage)
- Logic levels
  - V_{IH}, V_{IL}, V_{OH}, V_{OL}
  - Noise margin
- Propagation delay
- Transition time
- Power dissipation
- Fan-in and Fan-out
- Gate Cost

V_{CC} – Power Supply Voltage

- Common across a logic family (e.g., 5V for all HC parts)
- V_{CC} and GND commonly called "power supply rails"
  - Sometimes V_{DD} and V_{SS} for CMOS devices

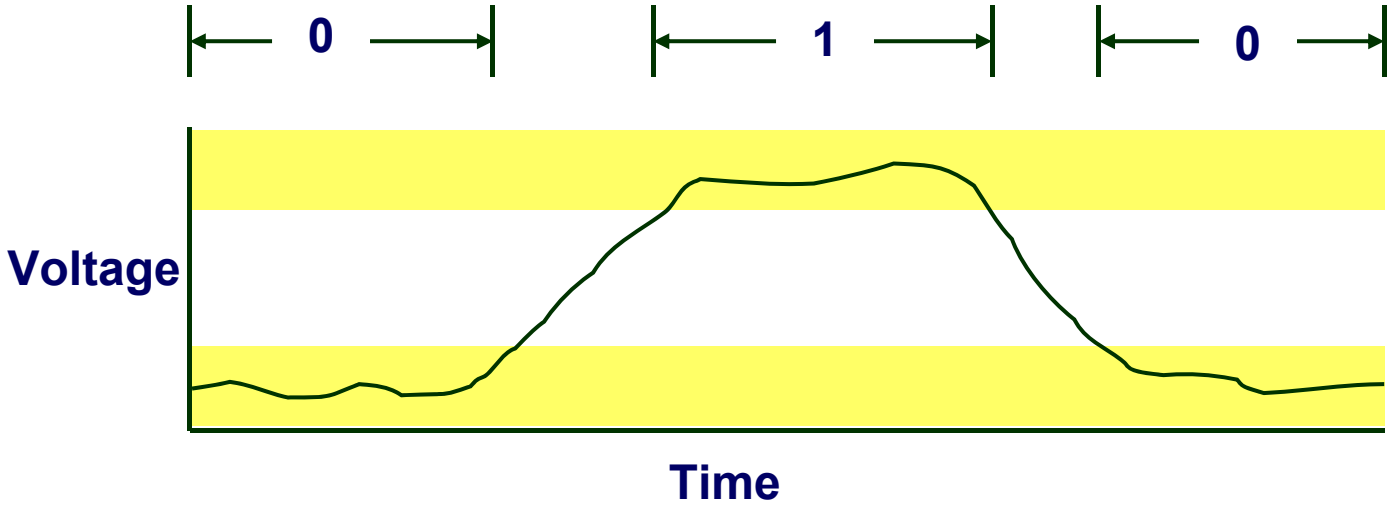


Logic Levels

- Discrete voltages represent 1 and 0
- For example:
  - 0 = ground (GND) or 0 volts
  - 1 = V_{CC} or 5 volts
- What about 4.99 volts? Is that a 0 or a 1?
- What about 3.2 volts?

- **Range** of voltages for 1 and 0
- Different ranges for inputs and outputs to allow for **noise**

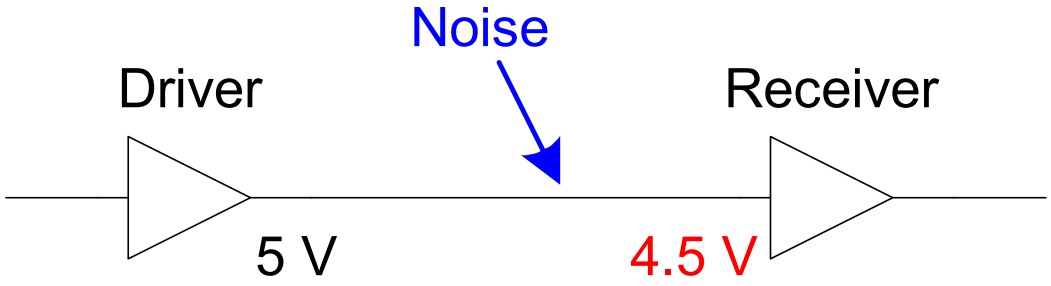
Use Voltage Thresholds to Extract Discrete Values from Continuous Signals



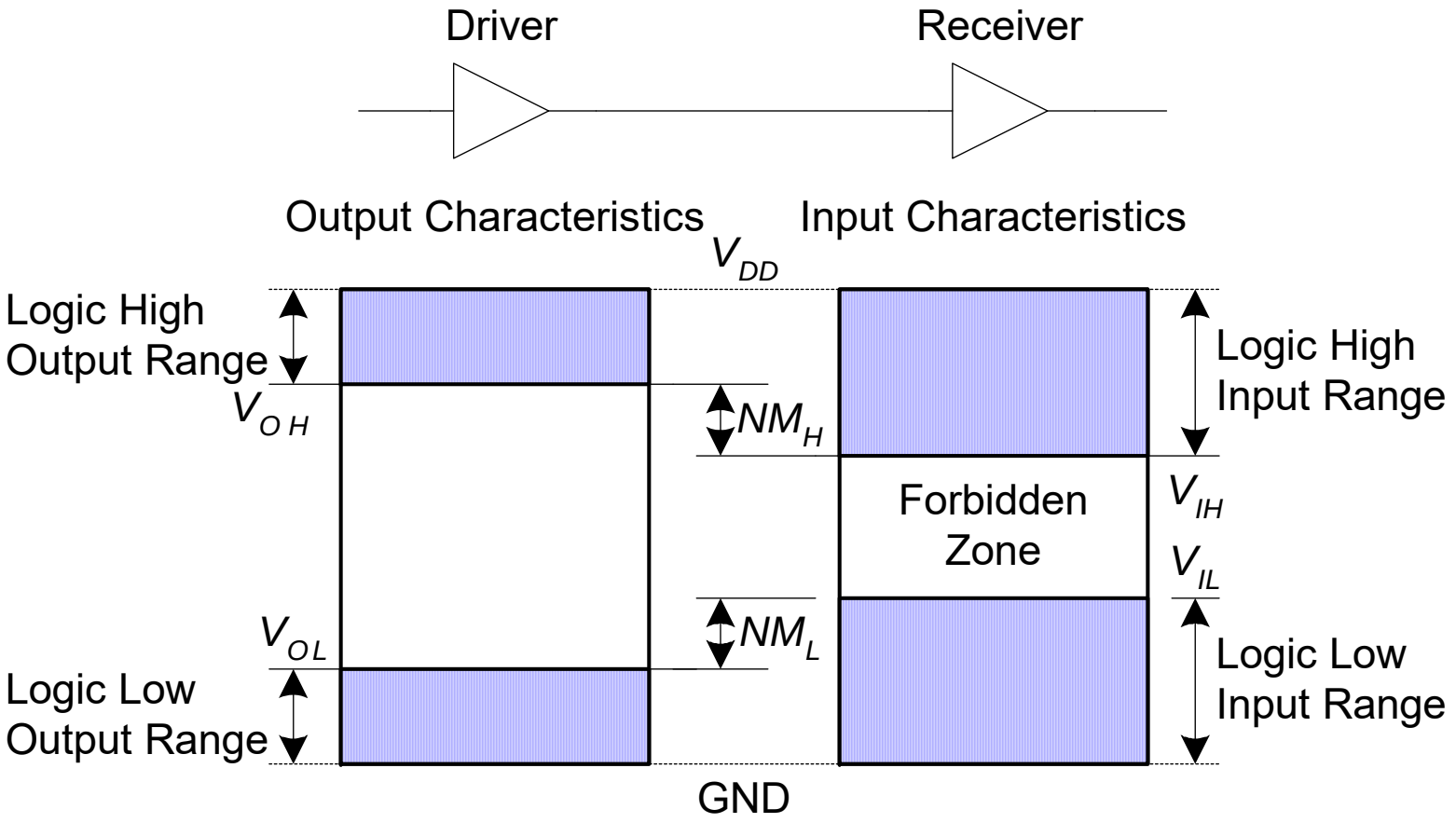
- Simplest version: 1-bit signal
  - Either high range (1) or low range (0)
  - With guard range between them
- Not strongly affected by noise or low-quality circuit elements
  - Can make circuits simple, small, and fast

What is Noise?

- **Anything that degrades the signal**
  - E.g., resistance, power supply noise, coupling to neighboring wires, etc.
- **Example**: a gate (driver) outputs 5 V but, because of resistance in a long wire, receiver gets 4.5 V



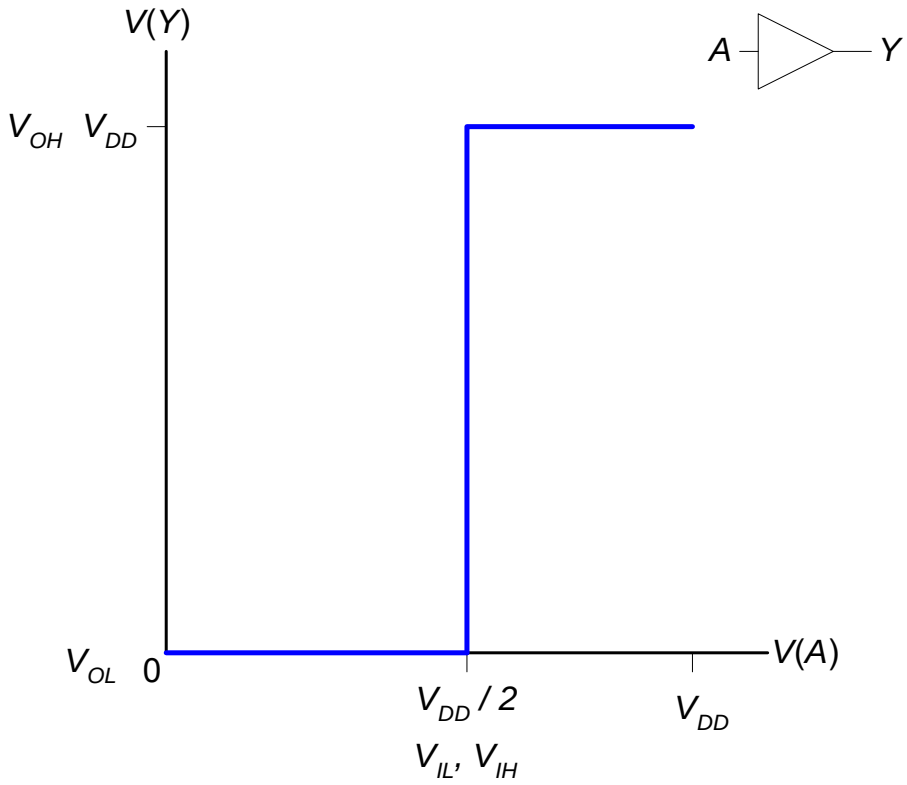
Noise Margins （以缓冲器为例）



$$\mathbf{NM_H = V_{OH} - V_{IH}}$$
$$\mathbf{NM_L = V_{IL} - V_{OL}}$$

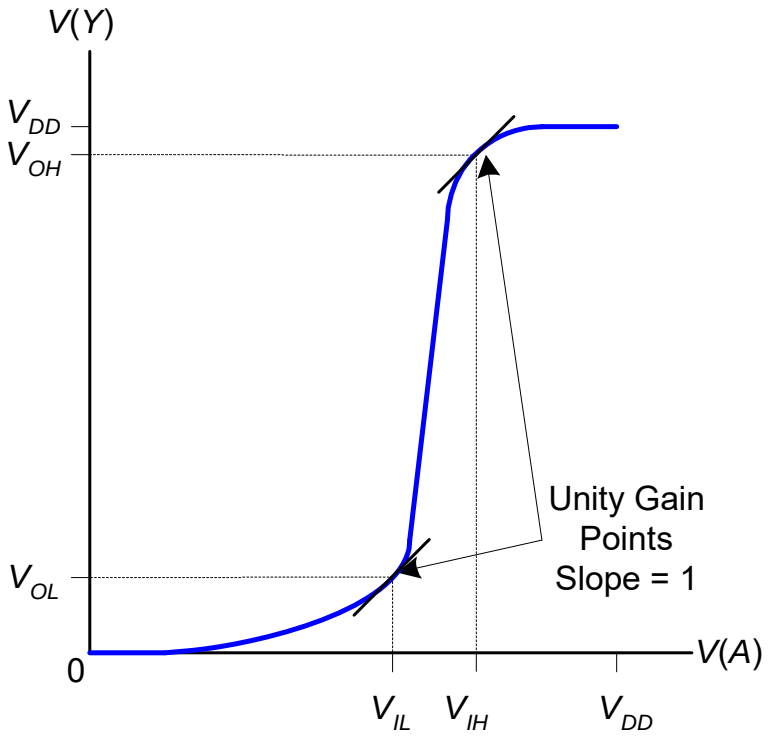
DC Transfer Characteristics

Ideal Buffer:



$$NM_H = NM_L = V_{DD} / 2$$

Real Buffer:



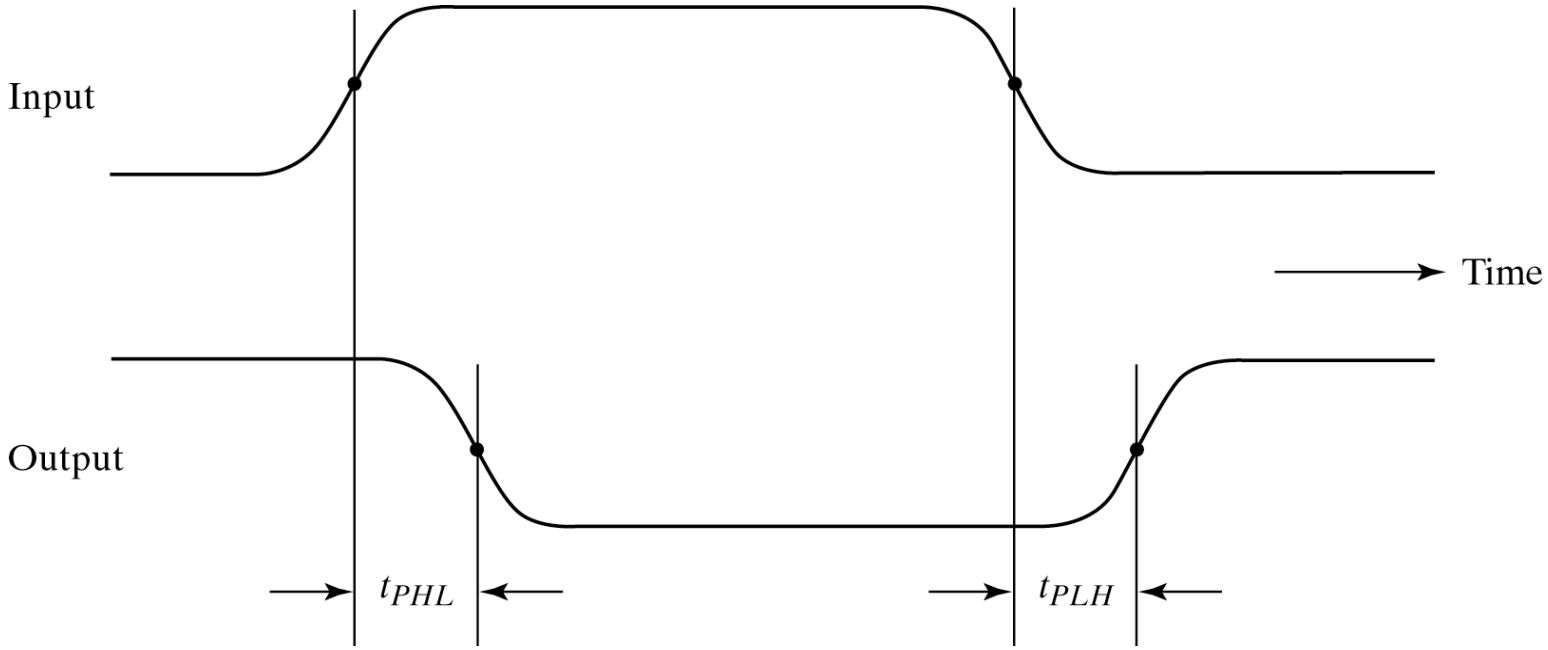
$$NM_H, NM_L < V_{DD} / 2$$

Logic Family Examples

Logic Family	V_{DD}	V_{IL}	V_{IH}	V_{OL}	V_{OH}
TTL	5 (4.75 - 5.25)	0.8	2.0	0.4	2.4
CMOS	5 (4.5 - 6)	1.35	3.15	0.33	3.84
LVTTL	3.3 (3 - 3.6)	0.8	2.0	0.4	2.4
LVCMOS	3.3 (3 - 3.6)	0.9	1.8	0.36	2.7

Propagation Delay (T_{pd}, T_{plh}, T_{phl})

- Outputs don't instantaneously reflect input changes
- Propagation delay is a measure of this time
- Delay from H to L can be different than from L to H

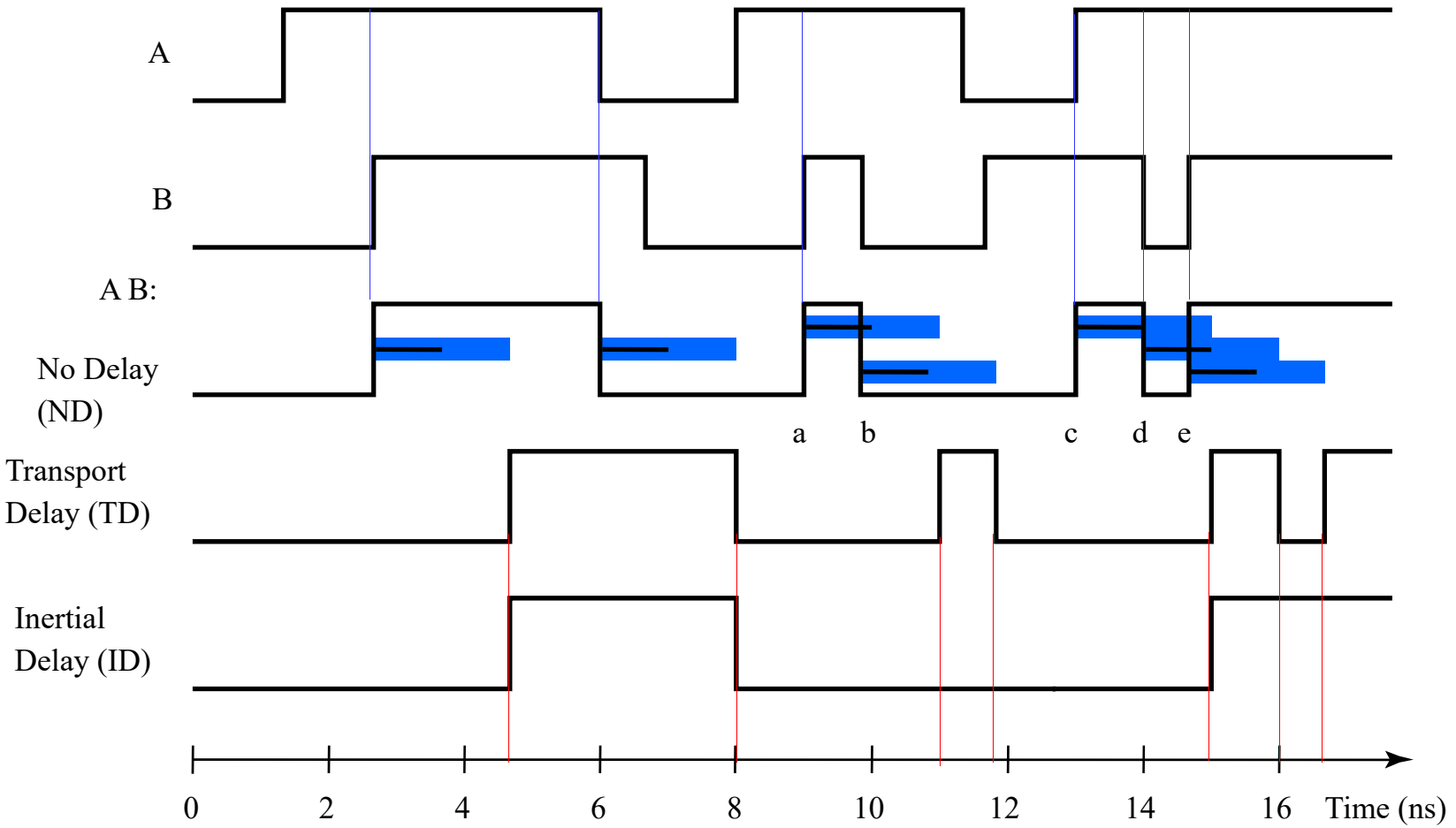


Delay Models

- *Transport delay* - a change of the output in response to a change on the inputs occurred after a fixed specified delay

- *Inertial delay* - similar to transport delay, except that if input changes cause the output to changes twice in an interval less than the *rejection time*, then the first of the output changes does not occur. This models typical electronic circuit behavior, namely, rejects narrow "pulses" on the outputs

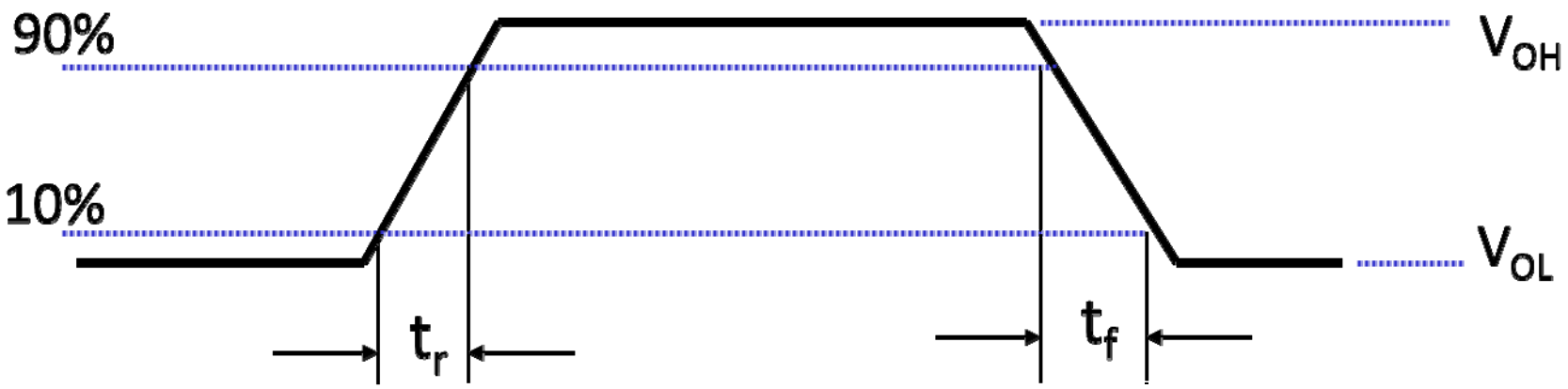
Delay Model Example



Propagation Delay = 2.0 ns **Rejection Time = 1.0 ns**

Transition Time (Rise & Fall)

- Takes time for signal to reach its output voltage
- Will be affected by capacitance, fan-out
- Also known as slew rate
- Typically measured from 10%/90% of V_{OL}/V_{OH} swing



Power Dissipation

- Power “dissipation” and “consumption”
- Static (“quiescent”) power dissipation
  - When device outputs not changing
  - Caused by the quiescent supply current, I_{DD} (also called the leakage current)： $P_S = I_{DD} \cdot V_{CC}$
  - Very low for CMOS
- Dynamic power dissipation

$$P_D = (C_{PD} + C_L) \cdot V_{CC}^2 \cdot f$$

C_{PD} = Power dissipation capacitance (constant for logic family)

C_L = Capacitive load on output (driving other devices)

V_{CC} = Supply voltage

f = Transition frequency (0.5 x number of output transitions/second)

Fan-in and Fan-out

- Fan-in refers to the maximum number of input signals that feed the input equations of a logic cell.
  - Fan-in is a term that defines the maximum number of digital inputs that a single logic gate can accept. Most transistor-transistor logic (TTL) gates have one or two inputs, although some have more than two. A typical logic gate has a fan-in of 1 or 2.
- The fan-out is defined as the maximum number of inputs (load) that can be connected to the output of a gate without degrading the normal operation.
  - Fan Out is calculated from the amount of current available in the output of a gate and the amount of current needed in each input of the connecting gate. It is specified by manufacturer and is provided in the data sheet. Exceeding the specified maximum load may cause a malfunction because the circuit will not be able supply the demanded power.

Fan-out

- Fan-out can be defined in terms of a standard load
  - Example: 1 standard load equals the load contributed by the input of 1 inverter.
  - *Transition time* -the time required for the gate output to change from H to L, t_{HL}, or from L to H, t_{LH}
  - The *maximum fan-out* that can be driven by a gate is the number of standard loads the gate can drive without exceeding its specified *maximum transition time*

Fan-out and Delay

- The fan-out loading of a gate's output affects the gate's propagation delay
- Example:
  - One equation for t_{pd} for a NAND gate with 4 inputs is:

    $$t_{pd} = 0.07 + 0.021 \text{ SL ns}$$

  - SL is the number of standard loads the gate is driving, i. e., its fan-out in standard loads
  - For SL = 4.5, $t_{pd} = 0.1645$ ns

Gate Cost

- In an integrated circuit:
  - The cost of a gate is proportional to the chip area occupied by the gate
  - The gate area is roughly proportional to the number and size of the transistors and the amount of wiring connecting them
  - Ignoring the wiring area, the gate area is roughly proportional to the gate input count
  - So gate input count is a rough measure of gate cost
- If the actual chip layout area occupied by the gate is known, it is a far more accurate measure

Further Reading (Optional)

编码

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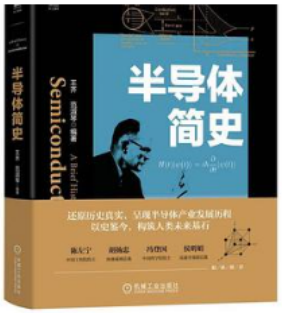
内容简介 · · · · · ·

本书讲述的是计算机工作原理。作者用丰富的想象和清晰的笔墨将看似繁杂的理论阐述得通俗易懂，你丝毫不会感到枯燥和生硬。更重要的是，你会因此而获得对计算机工作原理较深刻的理解。这种理解不是抽象层面上的，而是具有一定深度的。

Further Reading (Optional)

半导体简史



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内容简介 · · · · · ·

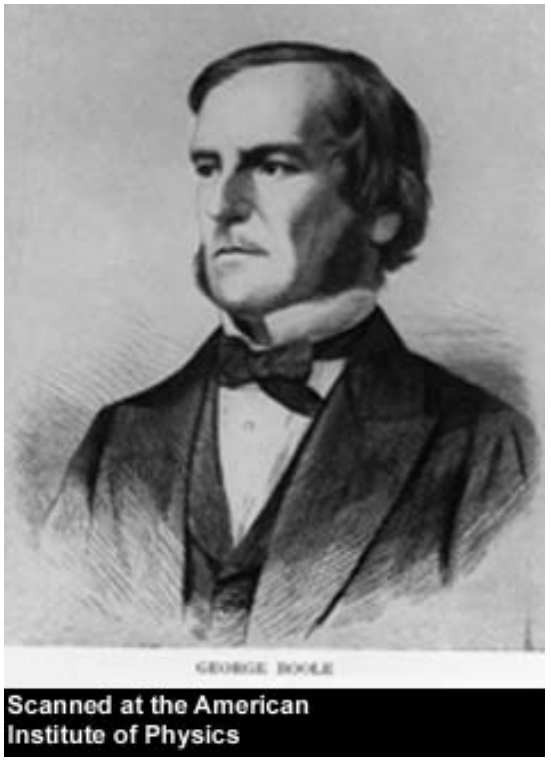
本书沿半导体全产业链的发展历程，分基础、应用与制造三条主线展开。其中，基础线主要覆盖与半导体材料相关的量子力学、凝聚态物理与光学的一些常识。应用线从晶体管与集成电路的起源开始，逐步过渡到半导体存储与通信领域。制造线以集成电路为主展开，并介绍了相应的半导体材料与设备。 三条主线涉及了大量与半导体产业相关的历史。笔者希望能够沿着历史的足迹，与读者一道在浮光掠影中领略半导体产业之全貌。 本书大部分内容以人物与公司传记为主，适用于绝大多数对半导体产业感兴趣的读者；部分内容涉及少许与半导体产业相关的材料理论，主要为有志于深入了解半导体产业的求职者或从业人员准备，多数读者可以将这些内容略去，并不会影响阅读的连续性。

Overview

- Introduction
- Binary logic and gates
- Transistors
- Some IC parameters
- Boolean algebra
- Logic functions
- Simplification of logic functions
- Additional Gates and Circuits

George Boole, 1815 - 1864

- Born to working class parents
- Taught himself mathematics and joined the faculty of Queen's College in Ireland.
- Wrote An Investigation of the Laws of Thought (1854)
- Introduced binary variables
- Introduced the three fundamental logic operations: AND, OR, and NOT.



Basic Concepts of Boolean Algebra

- **Boolean algebra is also known as logic algebra or switching algebra.**
  - **An algebra dealing with binary variables and logic operations**
  - **Fundamental in the development of computer science and digital logic**
- **Logic variables take on one of two values. We use 1 and 0 (or True and false) to denote the two values.**
  - **the value of variables is not comparable, and it only represents the state of an event.**
  - **For example, logic 1 may indicate switch is closed and logic 0 for switch open**
- **Three basic logic operations: AND, OR and NOT**

Definition of Boolean Function

Definition of Boolean Function!!!

Define that X1, X2, …, Xn are input signals of some logic circuit and F is the output signal. Once the input signals X1, X2, …, Xn are fixed, the output signal is fixed as well. F is called the Boolean function of X1, X2, …, Xn. The Boolean equation representing function F is

$$F=f(X1, X2, \ldots, Xn)$$

Note that: if we have X+Y=X+Z or XY=XZ, it doesn't mean we have Y=Z

Expression of Boolean Function

- *Boolean expression* is an algebraic expression formed by using binary variables, the constants 0 and 1, the logic operation symbols, and parentheses.

$$F(X,Y,Z) = X + \overline{Y}Z$$

  - A Boolean function can be described by no less than one Boolean equation
- There are Single-output Boolean function and Multiple-output Boolean function. And Multiple-output Boolean function has more than one function for the output signals.

❖ A Boolean function can also be described by truth table.

  - ❖ The truth table is unique!

X	Y	Z	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Table 2-2 Truth Table for the Function F

Some Properties & Identities of the Algebra

Basic Identities

For the Simplification of Boolean Expression

1.	$X+0=X$	2.	$X\cdot 1=X$	**0 - 1 law**
3.	$X+1=1$	4.	$X\cdot 0=0$	
5.	$X+X=X$	6.	$X\cdot X=X$	**Overlapping law**
7.	$X+\overline{X}=1$	8.	$X\cdot\overline{X}=0$	**Complementary law**
9.	$\overline{\overline{X}}=X$			**Involution law**
10.	$X+Y=Y+X$	11.	$XY=YX$	Commutative
12.	$X+(Y+Z)=(X+Y)+Z$	13.	$X(YZ)=(XY)Z$	Associative
14.	$X(Y+Z)=XY+XZ$	15.	$X+YZ=(X+Y)(X+Z)$	Distributive
16.	$\overline{X+Y}=\overline{X}\cdot\overline{Y}$	17.	$\overline{X\cdot Y}=\overline{X}+\overline{Y}$	DeMorgan's

$A(A+B)=A$	$A+AB=A$	**Absorptive law**
$A(\overline{A}+B)=AB$	$A+\overline{A}B=A+B$	**Including law**
$(A+B)(\overline{A}+C)(B+C)=(A+B)(\overline{A}+C)$	$AB+\overline{A}C+BC=AB+\overline{A}C$	

Theorems: n Variables

- Generalized Idempotent Theorem
  - (T12) $X + X + \ldots + X = X$
  - (T12D) $X \cdot X \cdot \ldots \cdot X = X$
- De Morgan's Theorem
  - (T13) $\overline{X_1 \cdot X_2 \cdot \ldots \cdot Xn} = \overline{X_1} + \overline{X_2} + \ldots + \overline{X_n}$
  - (T13D) $\overline{X_1 + X_2 + \ldots + Xn} = \overline{X_1} \cdot \overline{X_2} \cdot \ldots \cdot \overline{X_n}$
  - (T14) $\overline{F(X_1, X_2, \ldots, Xn, +, \cdot)} = F(\overline{X_1}, \overline{X_2}, \ldots, \overline{X_n}, \cdot, +)$
- Shannon's Theorem
  - (T15) $F(X_1, X_2, \ldots, X_n) = X_1 \cdot F(1, X_2, \ldots, X_n) + \overline{X_1} \cdot F(0, X_2, \ldots, X_n)$
  - (T15D) $F(X_1, X_2, \ldots, X_n) = [X_1 + F(0, X_2, \ldots, X_n) \cdot [\overline{X_1} + F(1, X_2, \ldots, X_n)]$

Equivalence of Boolean Function

For the following Boolean functions:

$$F_1 = f_1(X_1, X_2, \ldots, X_n)$$
$$F_2 = f_2(X_1, X_2, \ldots, X_n)$$

If **Any** input signal values of *X*1, *X*2, …, *Xn*, produces the **same** outputs *F1* and *F2*, Then we say that *F1* and *F2* are equal:

$F1 = F2$

Boolean Operator Precedence

- **The order of evaluation in a Boolean expression is:**

  1. Parentheses
  2. NOT
  3. AND
  4. OR

- **Consequence: Parentheses appear around OR expressions**
- **Example:** $F = A\,(B + C)\,(\overline{C + D})$
  - If the meaning is unambiguous, we leave out the symbol "."

Example 1: Boolean Algebraic Proof

- $A + A \cdot B = A$ (Absorption Theorem)

Proof Steps	Justification (identity or theorem)
$A + A \cdot B$	
$= A \cdot 1 + A \cdot B$	$X = X \cdot 1$
$= A \cdot (1 + B)$	$X \cdot Y + X \cdot Z = X \cdot (Y + Z)$(Distributive Law)
$= A \cdot 1$	$1 + X = 1$
$= A$	$X \cdot 1 = X$

- **Our primary reason for doing proofs is to learn:**
  - **Careful and efficient use of the identities and theorems of Boolean algebra, and**
  - **How to choose the appropriate identity or theorem to apply to make forward progress, irrespective of the application.**

Example 2: Boolean Algebraic Proofs

- **$AB + \overline{A}C + BC = AB + \overline{A}C$ (Consensus Theorem)**

Proof Steps

$AB + \overline{A}C + BC$

$= ?$

Example 3: Boolean Algebraic Proofs

- $(\overline{X + Y})Z + X\overline{Y} = \overline{Y}(X + Z)$

Proof Steps

$(\overline{X + Y})Z + X\overline{Y}$

$= ?$

Proof of DeMorgan's Laws

- $\overline{X + Y} = \overline{X} \cdot \overline{Y}$

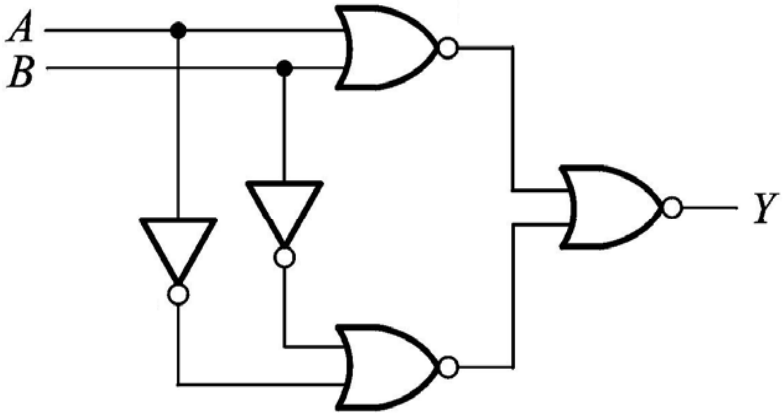
- $\overline{X \cdot Y} = \overline{X} + \overline{Y}$

Overview

- Introduction
- Binary logic and gates
- Digital abstraction
- Transistors
- Boolean algebra
- Logic functions
- Simplification of logic functions
- Additional Gates and Circuits

Representation of Logic Functions

- Logic Functions
  - $F(X_1, X_2, \ldots, X_n)$



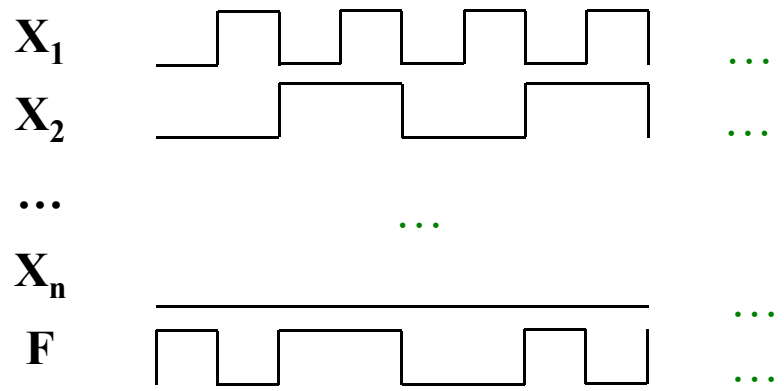
What is the logic function?

逻辑函数是反映输入变量和输出变量之间关系的逻辑关系表达式。

Truth Table vs. Waveforms

- Representations of Logic Functions
  - Truth Table
  - Waveforms

X_1	…	X_n	F
0	…	1	0
0	…	0	1
1	…	1	1
…	…	…	…
0	…	0	1



Truth Tables

- Truth table: a unique representation of a Boolean function
- If two functions have identical truth tables, the functions are equivalent (and vice-versa).
- Truth tables can be used to prove equality theorems.
- However, the size of a truth table grows exponentially with the number of variables involved, hence unwieldy. This motivates the use of Boolean Algebra.

Boolean expressions-NOT unique

- Unlike truth tables, expressions representing a Boolean function are NOT unique.
- Example:
- $F(x,y,z) = \bar{x}\bar{y}\bar{z} + \bar{x}y\bar{z} + xy\bar{z}$
- $G(x,y,z) = \bar{x}\bar{y}\bar{z} + y\bar{z}$
- The corresponding truth tables for F() and G() are to the right. They are identical.
- Thus, F() = G()

x	y	z		F	G
0	0	0		1	1
0	0	1		0	0
0	1	0		1	1
0	1	1		0	0
1	0	0		0	0
1	0	1		0	0
1	1	0		1	1
1	1	1		0	0

Complement of a Function

- The complement of a function is derived by interchanging (• and +), and (1 and 0), and **complementing each variable**.
- Otherwise, interchange 1s to 0s in the truth table column showing F.
- The complement of a function IS NOT THE SAME as the dual of a function.

Complementation: An Example

- Find G(x,y,z), the complement of
  $F(x,y,z) = x\bar{y}\bar{z} + \bar{x}yz$

- $G = F' = (x\bar{y}\bar{z} + \bar{x}yz)'$
  $= \overline{x\bar{y}\bar{z}} \bullet \overline{\bar{x}yz}$ *DeMorgan*
  $= (\bar{x} + y + z) \bullet (x + \bar{y} + \bar{z})$ *DeMorgan* again

- Note: The complement of a function can also be derived by finding the function's dual, and then complementing all of the literals

Some Properties of Identities & the Algebra

- The **dual** of an algebraic expression is obtained by interchanging + and · and interchanging 0's and 1's.
- The identities appear in dual pairs. When there is only one identity on a line the identity is self-dual, i.e., the dual expression = the original expression.
- Unless it happens to be self-dual, the dual of an expression does not equal the expression itself.
  - Example: $F = (A + \overline{C}) \cdot B + 0$
    - dual $F = (A \cdot \overline{C} + B) \cdot 1 = A \cdot \overline{C} + B$
  - Example: $G = X \cdot Y + \overline{(W + Z)}$
    - dual G =
  - Example: $H = A \cdot B + A \cdot C + B \cdot C$
    - dual H =
  - Are any of these functions self-dual?

Power of Duality

- $x + xy = x$ is true, so
  - $\overline{x + xy} = \bar{x}$
  - $\overline{x + xy} = \bar{x}\,(\bar{x} + \bar{y})$
  - $\bar{x}\,(\bar{x} + \bar{y}) = \bar{x}$
- Let $X = \bar{x}$, $Y = \bar{y}$
- $X(X+Y) = X$, which is the dual of $x + xy = x$.
- The above process can be applied to any formula. So if a formula is valid, then its dual must also be valid.
- Proving one formula also proves its dual!

Canonical and Standard Forms

- We need to consider formal techniques for the simplification of Boolean functions.
  - Identical functions will have exactly the same canonical form.
  - Minterms and Maxterms
  - Sum-of-Minterms and Product-of- Maxterms
  - Product and Sum terms
  - Sum-of-Products (SOP) and Product-of-Sums (POS)

Minterms

- **Minterms are AND terms with every variable present just once in either true or complemented form.**
- **Given that each binary variable may appear normal (e.g., x) or complemented (e.g., ~x), there are 2^n Minterms for n variables.**

Properties of Minterms:

1. **It represents exactly one combination in truth table**
2. **Product of any two Minterms is 0. $m_i \cdot m_j = 0, i \neq j$**
3. **Sum of all Minterms is 1. $\Sigma m_i = 1, i = 0 \sim 2^n - 1$**
4. **Each Minterm is either in function F or in the function's complement ~F**

Minterms for Three Variables

X	Y	Z	Product Term	Symbol	m_0	m_1	m_2	m_3	m_4	m_5	m_6	m_7
0	0	0	$\bar{X}\bar{Y}\bar{Z}$	m_0	1	0	0	0	0	0	0	0
0	0	1	$\bar{X}\bar{Y}Z$	m_1	0	1	0	0	0	0	0	0
0	1	0	$\bar{X}Y\bar{Z}$	m_2	0	0	1	0	0	0	0	0
0	1	1	$\bar{X}YZ$	m_3	0	0	0	1	0	0	0	0
1	0	0	$X\bar{Y}\bar{Z}$	m_4	0	0	0	0	1	0	0	0
1	0	1	$X\bar{Y}Z$	m_5	0	0	0	0	0	1	0	0
1	1	0	$XY\bar{Z}$	m_6	0	0	0	0	0	0	1	0
1	1	1	XYZ	m_7	0	0	0	0	0	0	0	1

Table 2-6: Minterms for Three Variables

Example:

$$F(A,B,C) = \overline{A}B\overline{C} + \overline{A}BC + AB\overline{C} + ABC$$

$$F(A,B,C) = \overline{A}B\overline{C} + \overline{A}BC + AB\overline{C} + ABC$$

$$= m2 + m3 + m6 + m7$$

$$= \Sigma\, m^{3}(2, 3, 6, 7)$$

Since: $f(A_1, A_2, \cdots, A_n) + \overline{f}(A_1, A_2, \cdots, A_n) = 1$

And $f(A_1, A_2, \cdots, A_n) + \overline{f}(A_1, A_2, \cdots, A_n) = \sum_{i=0}^{2^n-1} m_i$

Thus $\sum_{i=0}^{2^n-1} m_i = 1$

Confirm ***Property 3***

Maxterms

- **Maxterms are OR terms with every variable appearing just once in true or complemented form.**
- **Given that each binary variable may appear normal (e.g., x) or complemented (e.g., ~x), there are 2^n maxterms for n variables.**

Properties of Maxterms:

1) **only one combination in truth table produces 0 result**
2) **Sum of any two Maxterms is 1. Mi+Mj=1, i≠j**
3) **Product of all Maxterms is 0.** $\prod_{i=0}^{2^n-1} M_i = 0$
4) **Each Maxterm is either in function F or in the function's complement ~F**

Maxterms for Three Variables

X	Y	Z	Sum Term	Symbol	M_0	M_1	M_2	M_3	M_4	M_5	M_6	M_7
0	0	0	$X+Y+Z$	M_0	0	1	1	1	1	1	1	1
0	0	1	$X+Y+\overline{Z}$	M_1	1	0	1	1	1	1	1	1
0	1	0	$X+\overline{Y}+Z$	M_2	1	1	0	1	1	1	1	1
0	1	1	$X+\overline{Y}+\overline{Z}$	M_3	1	1	1	0	1	1	1	1
1	0	0	$\overline{X}+Y+Z$	M_4	1	1	1	1	0	1	1	1
1	0	1	$\overline{X}+Y+\overline{Z}$	M_5	1	1	1	1	1	0	1	1
1	1	0	$\overline{X}+\overline{Y}+Z$	M_6	1	1	1	1	1	1	0	1
1	1	1	$\overline{X}+\overline{Y}+\overline{Z}$	M_7	1	1	1	1	1	1	1	0

Table 2-7: Maxterms for Three Variables

Maxterms and Minterms

1) M_i and m_i are complementary for each other.

$$M_i = \overline{m}_i \text{ and } m_i = \overline{M}_i$$

Example: $m_3 = \overline{A}\ B\ C$ and $M_3 = A + \overline{B} + \overline{C}$

2) Boolean Function $F = \Sigma m_i = \overline{\Pi M_i}$

Example: Describe Minterms and Maxterms with Truth Table

X	Y	Z	F	$\overline{F}$
0	0	0	1	0
0	0	1	0	1
0	1	0	1	0
0	1	1	0	1
1	0	0	0	1
1	0	1	1	0
1	1	0	0	1
1	1	1	1	0

Minterms

$$F_{\min} = \overline{X}\,\overline{Y}\,\overline{Z} + \overline{X}Y\overline{Z} + X\overline{Y}Z + XYZ$$

Maxterms

$$F_{\max} = \left(X + Y + \overline{Z}\right)\left(X + \overline{Y} + \overline{Z}\right)\left(\overline{X} + Y + Z\right)\left(\overline{X} + \overline{Y} + Z\right)$$

$$F_{\min}\left(X, Y, Z\right) = m_0 + m_2 + m_5 + m_7 = \sum\left(0, 2, 5, 7\right)$$

$$F_{\max}\left(X, Y, Z\right) = M_1 \bullet M_3 \bullet M_4 \bullet M_6 = \prod\left(1, 3, 4, 6\right)$$

$$\overline{F_{\min}\left(X, Y, Z\right)} = \overline{X}\,\overline{Y}Z + \overline{X}YZ + X\overline{Y}\,\overline{Z} + XY\overline{Z}$$

$$= m_1 + m_3 + m_4 + m_6 = \sum\left(1, 3, 4, 6\right)$$

Standard Forms

- **Sum-of-Products (SOP) form: equations are written as an OR of AND terms**
- **Product-of-Sums (POS) form: equations are written as an AND of OR terms**
- **Examples:**
  - **SOP:** $A\ B\ C + \overline{A}\ \overline{B}\ C + B$
  - **POS:** $(A + B) \cdot (A + \overline{B} + \overline{C}) \cdot C$
- **These "mixed" forms are neither SOP nor POS**
  - $(A\ B + C)\ (A + C)$
  - $A\ B\ \overline{C} + A\ C\ (A + B)$

Sum-of-Products (SOP)

- **A Simplification Example:**
- **$F(A,B,C) = \Sigma m(1,4,5,6,7)$**
- **Writing the minterm expression:**

  $F = A'B'C + AB'C' + AB'C + ABC' + ABC$
- **Simplifying:**

  $F = A'B'C + AB'(C' + C) + AB(C' + C)$

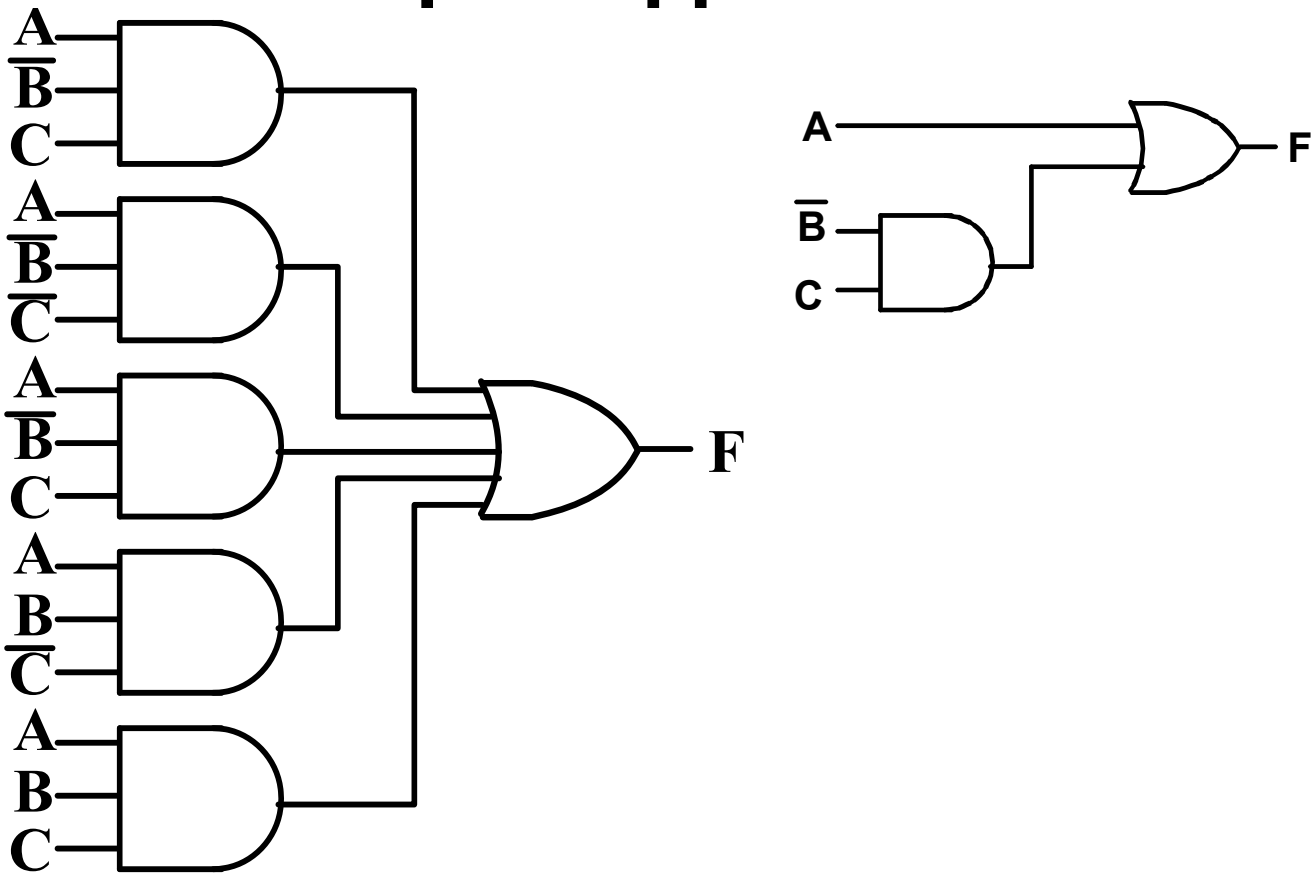
  $= A'B'C + AB' + AB$

  $= A'B'C + A$

  $= B'C + A$
- **Simplified F contains 3 literals compared to 15 in minterm F**

Two-level Implementation of SOP Expression

- **The two implementations for F are shown below – it is quite apparent which is**



SOP and POS Observations

- **The previous examples show that:**
  - **Standard Forms (Sum-of-minterms, Product-of-Maxterms), or other standard forms (SOP, POS) differ in complexity**
  - **Boolean algebra can be used to manipulate equations into simpler forms.**
  - **Simpler equations lead to simpler two-level implementations**
- **Questions:**
  - **How can we attain a "simplest" expression?**
  - **Is there only one minimum cost circuit?**
  - **The next part will deal with these issues.**

Overview

- Introduction
- Binary logic and gates
- Transistors
- Some IC parameters
- Boolean algebra
- Logic functions
- Simplification of logic functions
- Additional Gates and Circuits

Simplification of Logic Functions

- About Cost Criterion
- Algebra Method
  - Use the theorems
- Karnaugh Map
  - Graphical representations

Circuit Optimization

- **Goal: To obtain the simplest implementation for a given function**
  - Optimization is a more formal approach to simplification that is performed using a specific procedure or algorithm
- **Optimization requires a cost criterion to measure the simplicity of a circuit**
- **Distinct cost criteria we will use:**
  - **Literal cost (L)**
  - **Gate input cost (G)**
  - **Gate input cost with NOTs (GN)**

Literal Cost

- **Literal**
  - **A variable or its complement**
- **Literal cost**
  - **The number of literal appearances in a Boolean expression corresponding to the logic circuit diagram**
- **Examples:**
  - $F = BD + A\overline{B}C + A\overline{C}\overline{D}$ **L = 8**
  - $F = BD + A\overline{B}C + A\overline{B}\overline{D} + AB\overline{C}$ **L =**
  - $F = (A + B)(A + D)(B + C + \overline{D})(\overline{B} + \overline{C} + D)$ **L =**
  - **Which solution is best?**

Literal Cost

- **Another Example:**
  - $F = ABCD + \overline{A}\overline{B}\overline{C}\overline{D}$
  - $F = (A + B)(B + \overline{C})(C + \overline{D})(D + \overline{A})$
  - **Which solution is best?**

Gate Input Cost

- **Gate input costs**
  - **The number of inputs to the gates in the implementation corresponding exactly to the given equation or equations.**
  - **G - inverters not counted, GN - inverters counted**
- **For SOP and POS equations, it can be found from the equation(s) by finding the sum of:**
  - **All literal appearances**
  - **The number of terms excluding single literal terms,(G) and**
  - **Optionally, the number of distinct complemented single literals (GN).**
- **Example:**
  - $F = BD + A\overline{B}C + A\overline{C}\overline{D}$ **L = 8, G = 11, GN = 14**
  - $F = BD + A\overline{B}C + A\overline{B}\overline{D} + AB\overline{C}$ **L = 11, G = ?, GN = ?**
  - $F = (A + B)(A + D)(B + C + \overline{D})(\overline{B} + \overline{C} + D)$ **L = 10, G = ?, GN = ?**
  - **Which solution is best?**

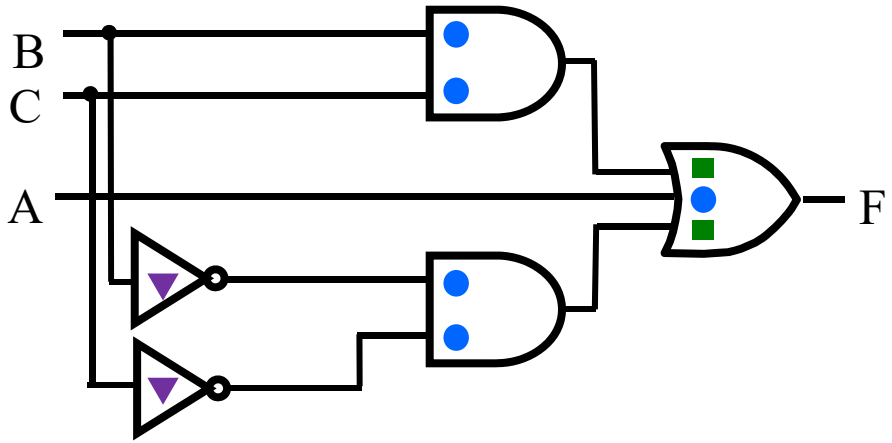
Cost Criteria (continued)

- **Example 1:**
- $F = A + BC + \overline{B}\,\overline{C}$

GN = G + 2 = 9

L = 5

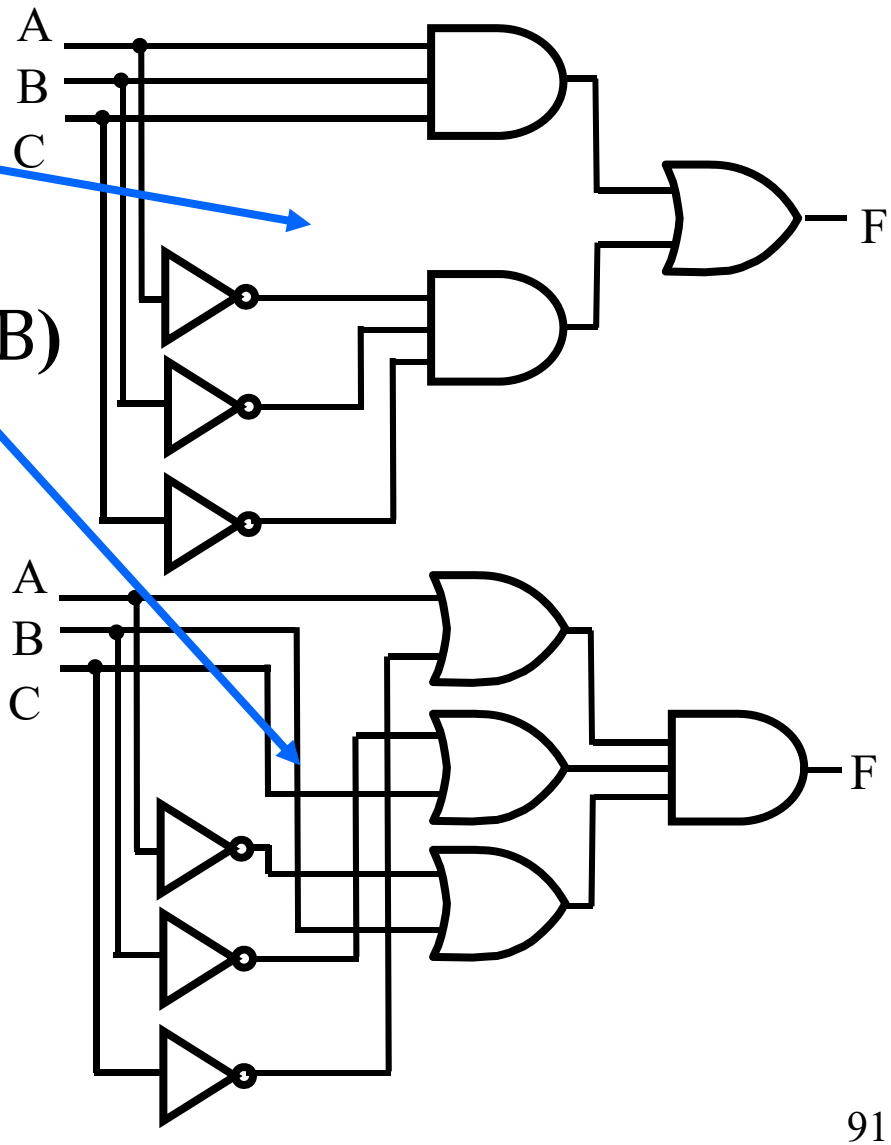
G = L + 2 = 7



- **L (literal count) counts the AND inputs and the single literal OR input**
- **G (gate input count) adds the remaining OR gate inputs**
- **GN(gate input count with NOTs) adds the inverter inputs**

Cost Criteria (continued)

- **Example 2:**
- $F = ABC + \overline{A}\overline{B}\overline{C}$
  - **L = 6 G = 8 GN = 11**
- $F = (A + \overline{C})(\overline{B} + C)(\overline{A} + B)$
  - **L = 6 G = 9 GN = 12**
- **<u>Same</u> function and <u>same</u> literal cost**
- **But first circuit has <u>better</u> gate input count and <u>better</u> gate input count with NOTs**
- **Select it!**



Boolean Function Optimization

- **Minimizing the gate input (or literal) cost of a (a set of) Boolean equation(s) reduces circuit cost.**
- **Boolean Algebra and graphical techniques are tools to minimize cost criteria values.**
- **Treat optimum or near-optimum cost functions for two-level (SOP and POS) circuits first.**
- **Introduce a graphical technique using Karnaugh maps (K-maps, for short)**

Algebraic Manipulation

- Boolean algebra is a useful tool for simplifying digital circuits.
- Why do it? Simpler can mean cheaper, smaller, faster.
- Example: Simplify $F = \bar{x}yz + \bar{x}y\bar{z} + xz$.

$$
\begin{aligned}
F &= \bar{x}yz + \bar{x}y\bar{z} + xz \\
&= \bar{x}y(z+\bar{z}) + xz \\
&= \bar{x}y \bullet 1 + xz \\
&= \bar{x}y + xz
\end{aligned}
$$

Yet Another One

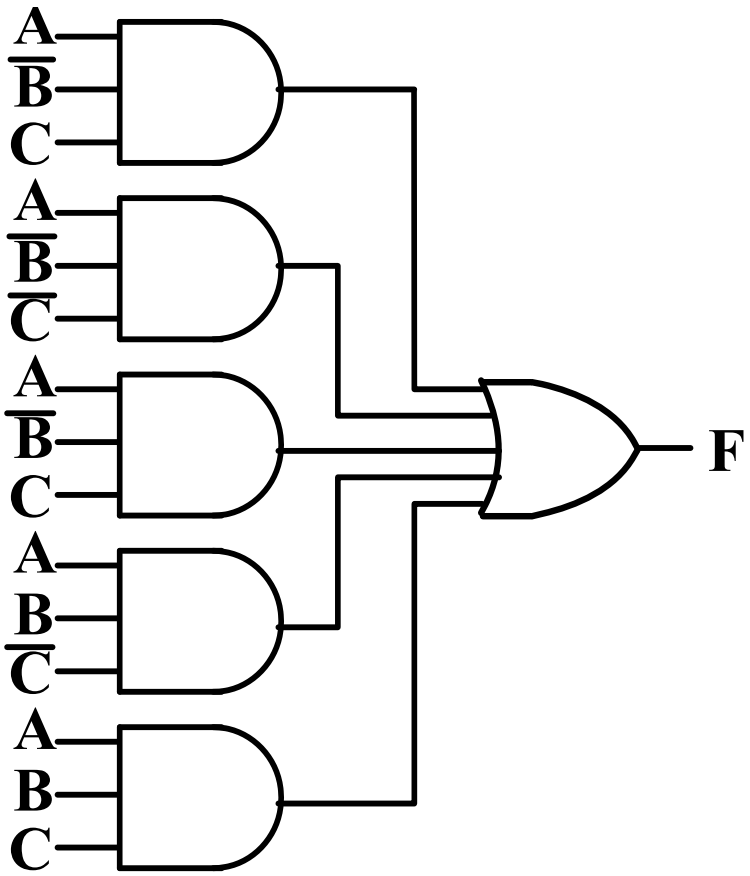
- A Simplification Example:
- **$F(A, B, C) = \Sigma m(1,4,5,6,7)$**
- Writing the minterm expression:

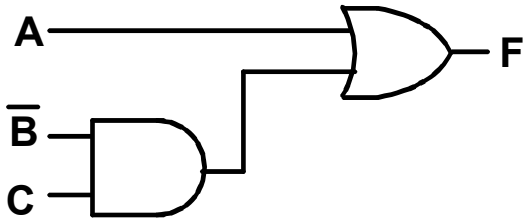
  $F = \overline{A}\,\overline{B}\,C + A\,\overline{B}\,\overline{C} + A\,\overline{B}\,C + AB\overline{C} + ABC$
- Simplifying:

  F =

The Benefit

- **The two implementations for F are shown below – it is quite apparent which is simpler!**



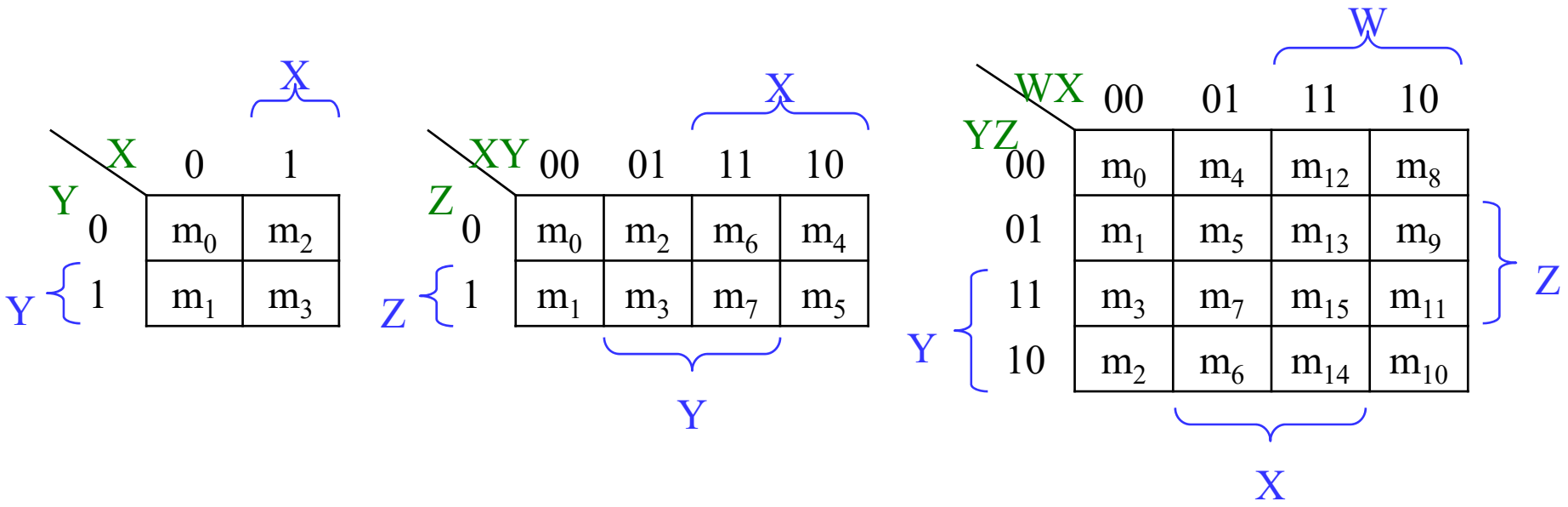


Karnaugh Maps (K-map)

- A K-map is a collection of squares (2K). And each square represents a Minterm. Adjacent squares differ in the value of one variable.
- Karnaugh maps (K-maps) are graphical representations of boolean functions.
- One **map cell** corresponds to a row in the truth table.
- Also, one map cell corresponds to a minterm or a maxterm in the boolean expression
- Multiple-cell areas of the map correspond to standard terms.

N-Variable Karnaugh Map

- Matrix with 2^n cells
  - E.g., 2-variables, 3-variables and 4-variables



Some Uses of K-Maps

- Finding optimum or near optimum
  - SOP and POS standard forms, and
  - two-level AND/OR and OR/AND circuit implementations

  for functions with small numbers of variables
- Visualizing concepts related to manipulating Boolean expressions, and
- Demonstrating concepts used by computer-aided design programs to simplify large circuits

Principles of Optimization with K-Map

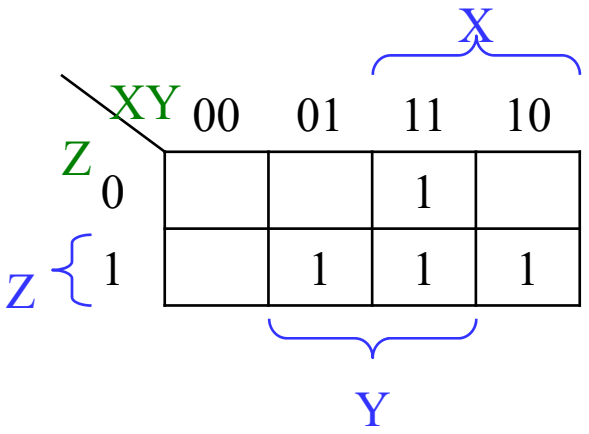
- Since $AB+A\overline{B}=A$, two adjacent minterms can be merged into one term to eliminate the redundant variable.
- In K-Map, the adjacent squares (each square refer to one minterm in the Boolean function) can be combined in similar way. And the combined squares are called Karnaugh Circles.
- Finding adjacent squares is very important in simplification with K-Map.

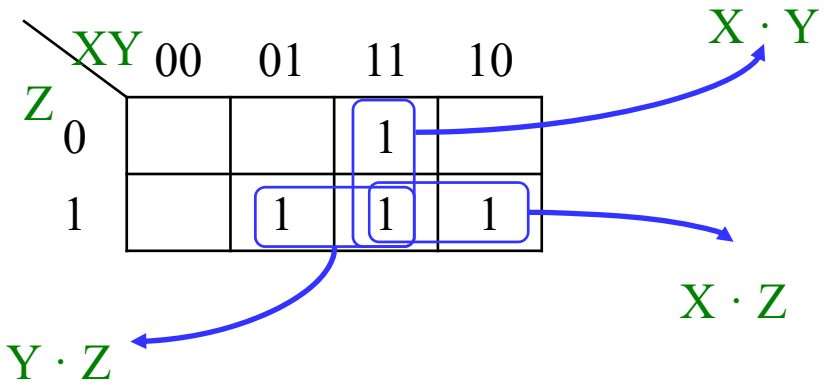
❖ The number of adjacent squares in Karnaugh Circles is a power of 2.

Simplification of Logic Functions

- Example: $F(X, Y, Z) = \sum m(3, 5, 6, 7)$

Index	X Y Z	F(X, Y, Z)
0	0 0 0	0
1	0 0 1	0
2	0 1 0	0
3	0 1 1	1
4	1 0 0	0
5	1 0 1	1
6	1 1 0	1
7	1 1 1	1





K-Map Definitions

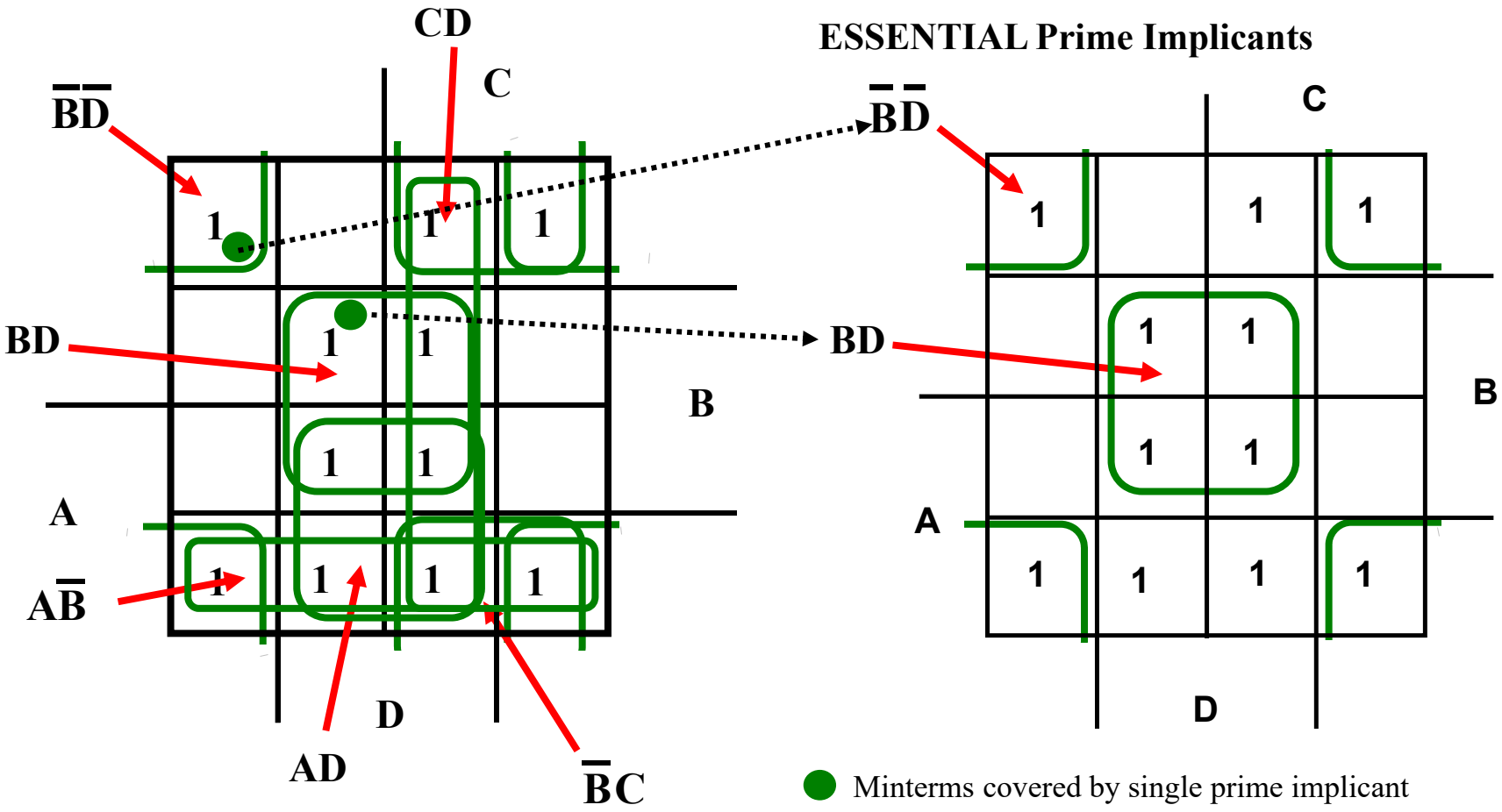
- **Implicant**
  - Product of literals $A\bar{B}C$, $\bar{A}C$, BC
- **Prime implicant**
  - Implicant corresponding to the largest circle in a K-map, i.e., is a product term obtained by combining the maximum possible number of adjacent squares in the map into a rectangle with the number of squares a power of 2.
- **Essential prime implicant**
  - A prime implicant is called an Essential Prime Implicant if it is the only prime implicant that covers (includes) one or more minterms.

K-Map Rules

- Every 1 must be circled at least once
- Each circle must span a power of 2 (i.e., 1, 2, 4) squares in each direction
- Each circle must be as large as possible
- A circle may wrap around the edges
- A “don't care” (X) is circled only if it helps minimize the equation

Example of Prime Implicants

- Find ALL Prime Implicants



Prime Implicant Practice

- **Find all prime implicants for:**

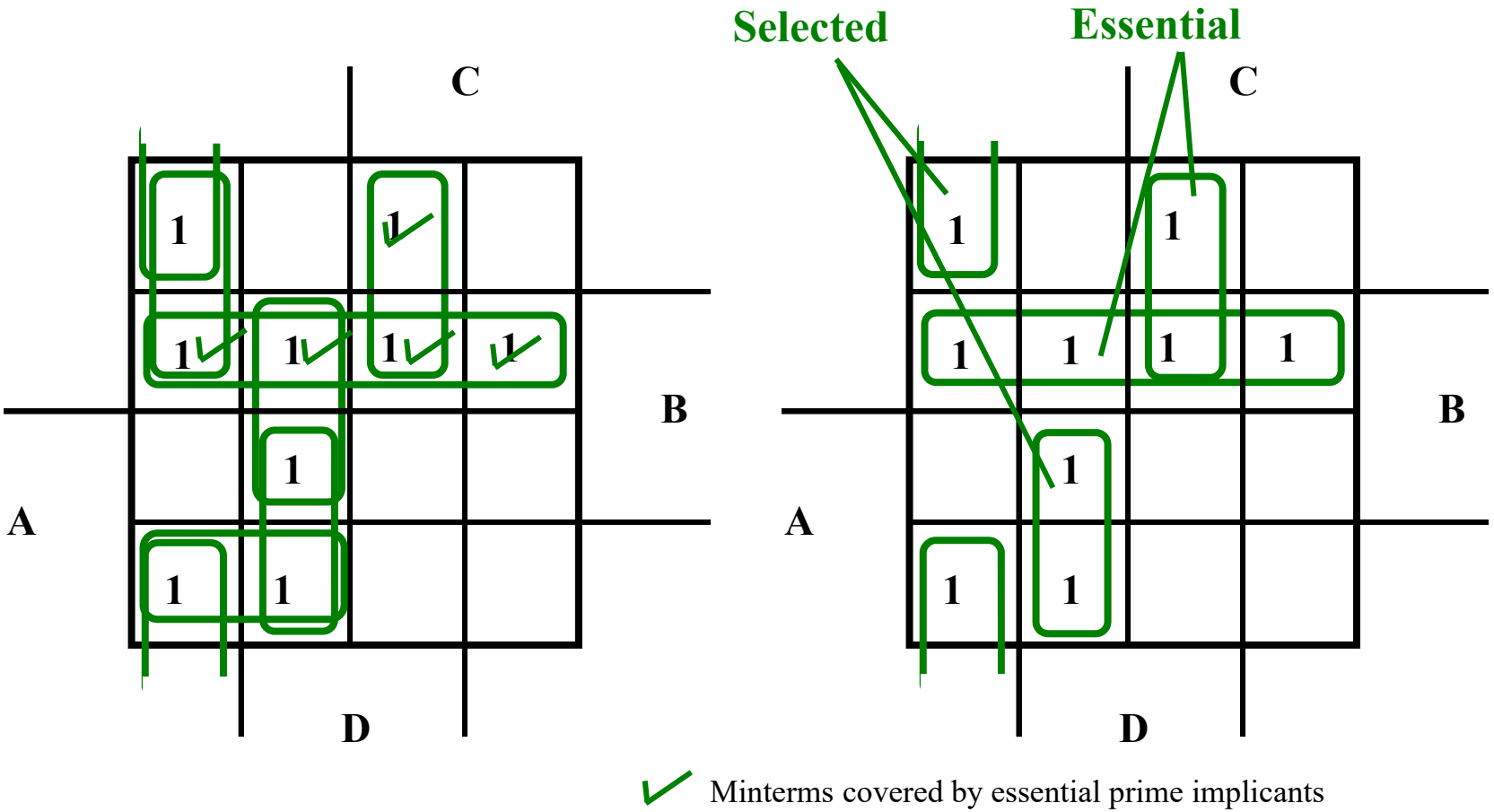
$$F(A,B,C,D) = \Sigma_m(0,2,3,8,9,10,11,12,13,14,15)$$

Prime Implicant Selection Rule

- **Minimize the overlap among prime implicants as much as possible.**
- **In particular, in the final solution, make sure that each prime implicant selected includes at least one minterm not included in any other prime implicant selected.**

Selection Rule Example

- Simplify F(A, B, C, D) given on the K-map



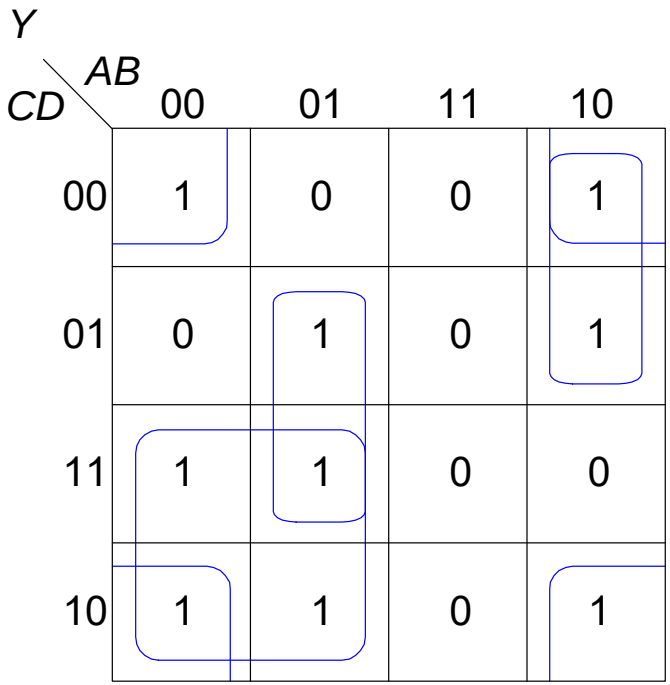
An Example of 4-Variable K-Map

A	*B*	*C*	*D*	*Y*
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

Y / *CD* \ *AB*	00	01	11	10
00	1	0	0	1
01	0	1	0	1
11	1	1	0	0
10	1	1	0	1

An Example of 4-Variable K-Map

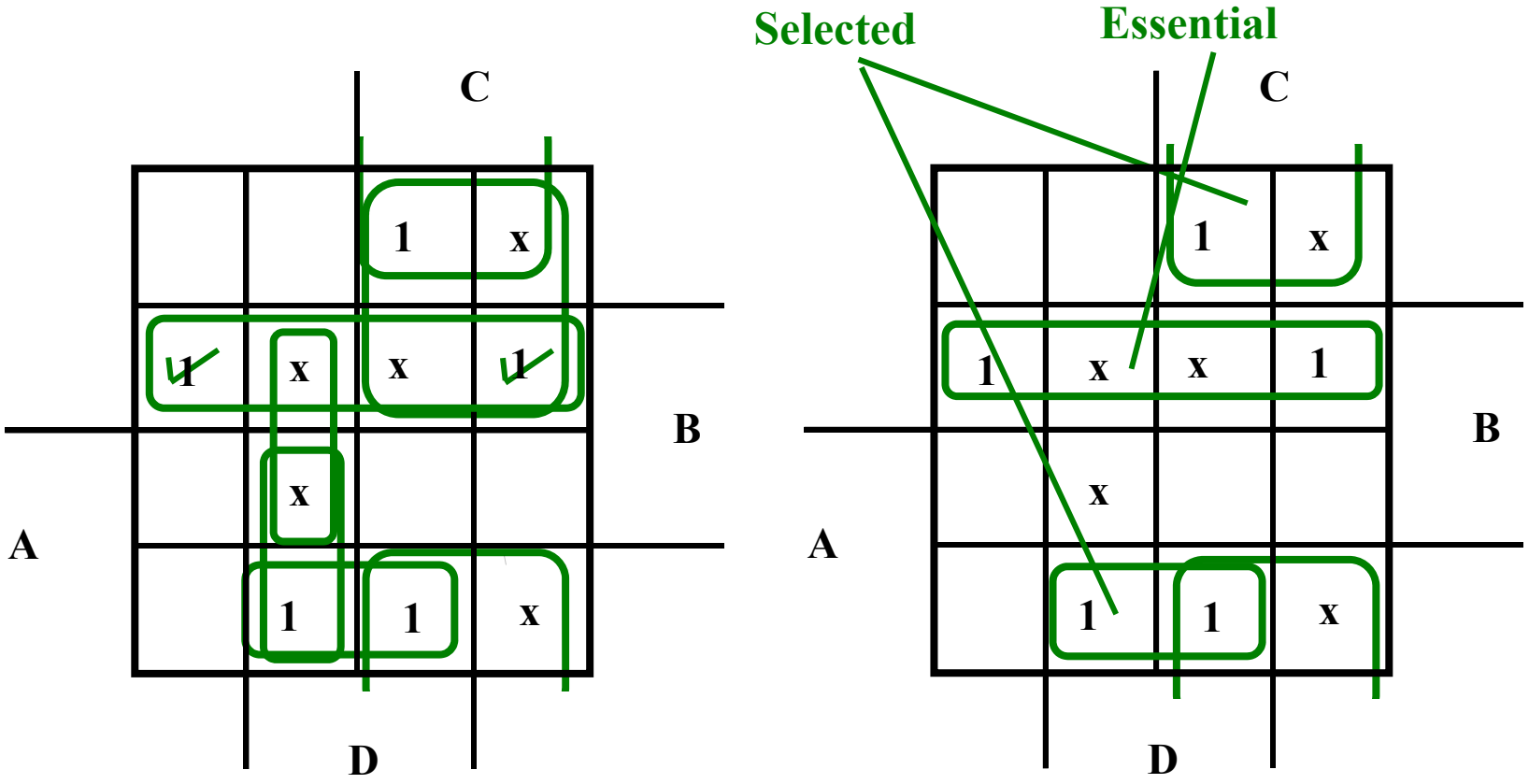
A	B	C	D	Y
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0



$$Y = \overline{A}C + \overline{A}BD + A\overline{B}\,\overline{C} + \overline{B}\,\overline{D}$$

Selection Rule Example with Don't Cares

- Simplify F(A, B, C, D) given on the K-map



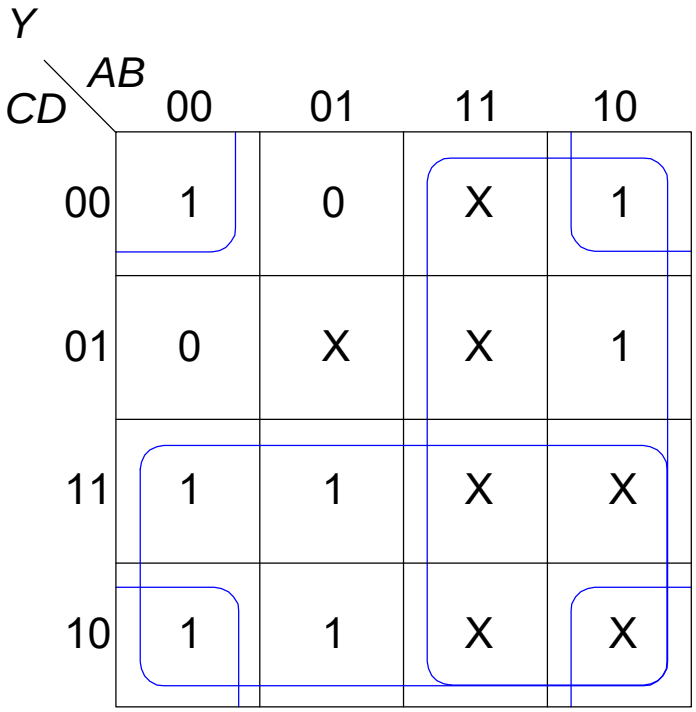
An Example with Don’t Cares

A	*B*	*C*	*D*	*Y*
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	X
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

Y	*AB* 00	01	11	10
CD 00	1	0	X	1
01	0	X	X	1
11	1	1	X	X
10	1	1	X	X

An Example with Don't Cares

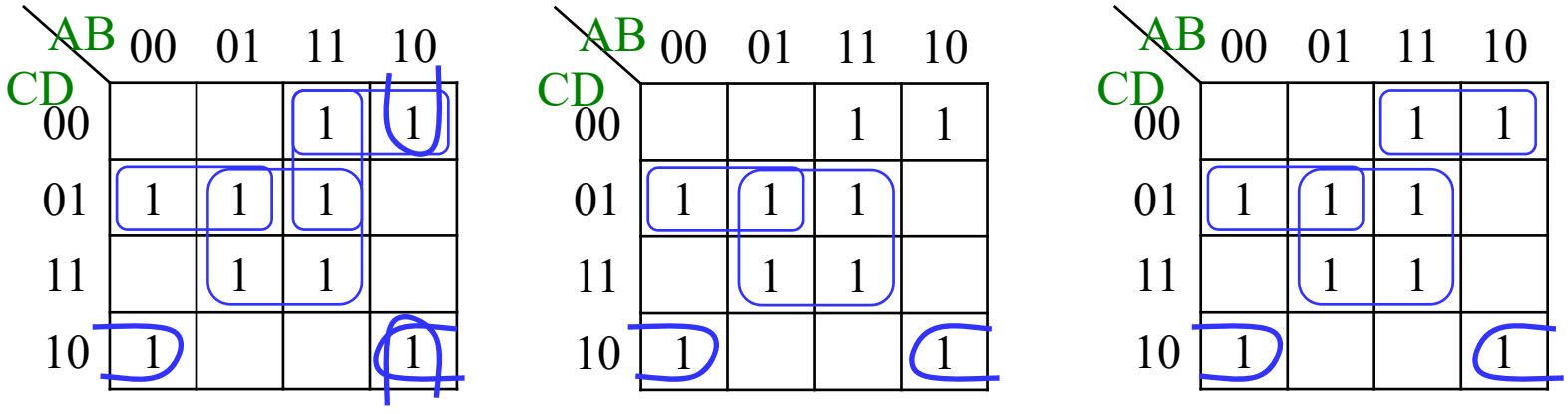
A	B	C	D	Y
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	X
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X



$Y = A + \overline{B}\,\overline{D} + C$

Systematic Simplification

- Example: $\sum_m(1, 2, 5, 7, 8, 10, 12, 13, 15)$



Systematic Simplification

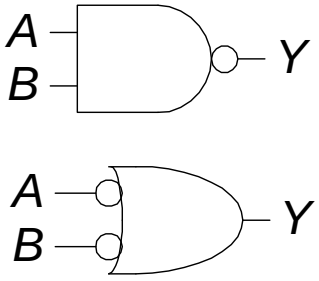
- $F(W,X,Y,Z) = \sum_m(3, 4, 5, 7, 9, 13, 14, 15)$?

Results May NOT Be Unique

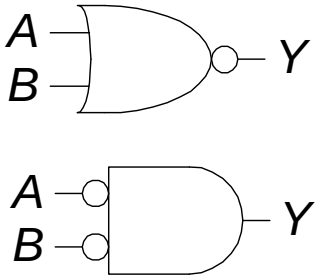
- $F(X, Y, Z) = XZ' + X'Z + YZ' + Y'Z$
- $F(X, Y, Z) = \sum_m$?
- ?

DeMorgan's Theorem Can Be Useful

- $Y = \overline{AB} = \overline{A} + \overline{B}$

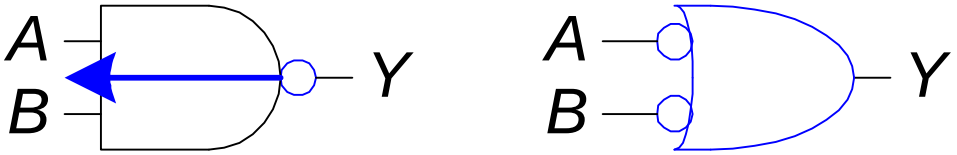


- $Y = \overline{A + B} = \overline{A} \bullet \overline{B}$

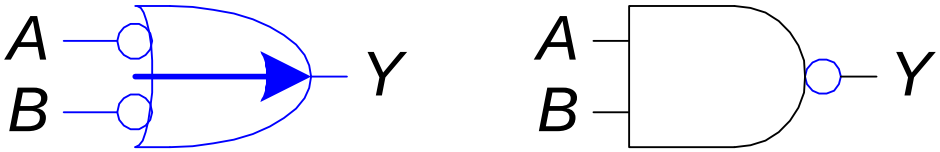


Bubble Pushing

- **Backward:**
  - Body changes
  - Adds bubbles to inputs

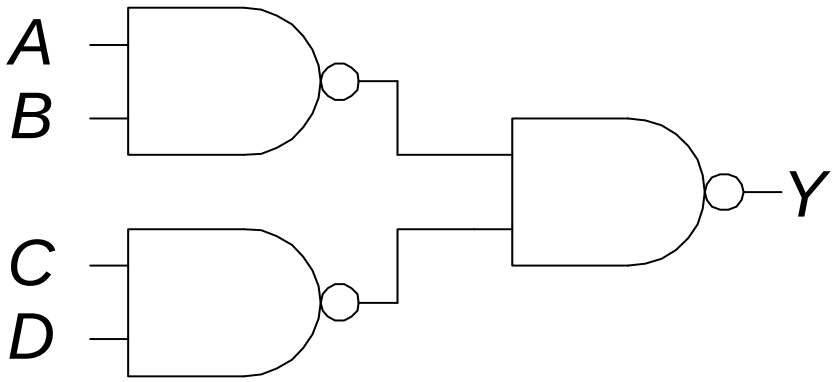


- **Forward:**
  - Body changes
  - Adds bubble to output



Bubble Pushing

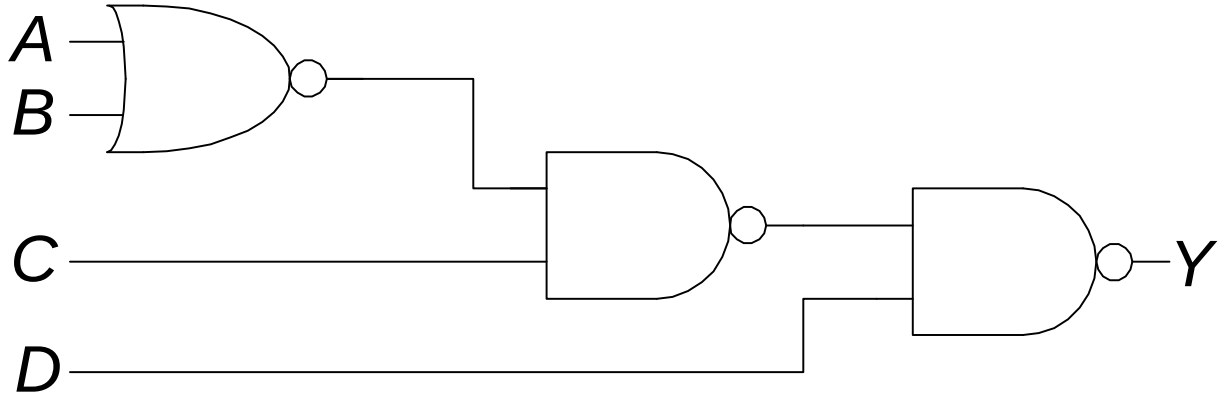
- What is the Boolean expression for this circuit?



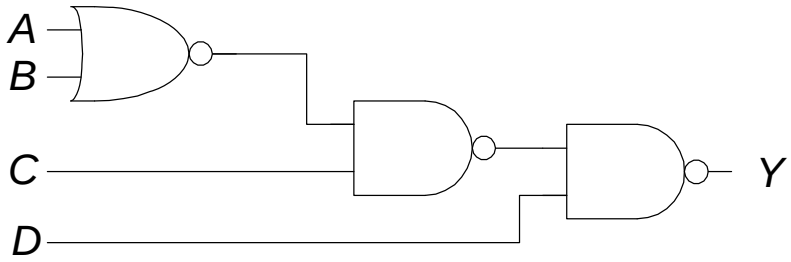
$$Y = AB + CD$$

Bubble Pushing Rules

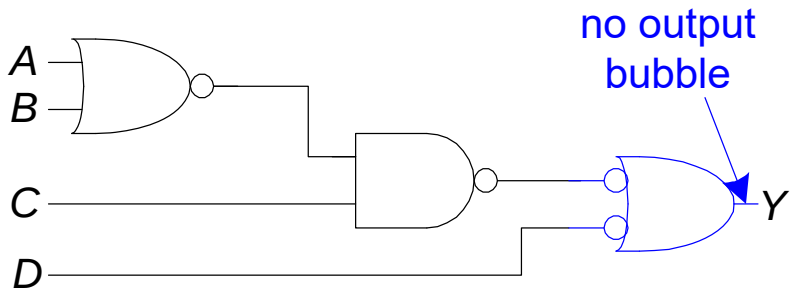
- Begin at output, then work toward inputs
- Push bubbles on final output back
- Draw gates in a form so bubbles cancel



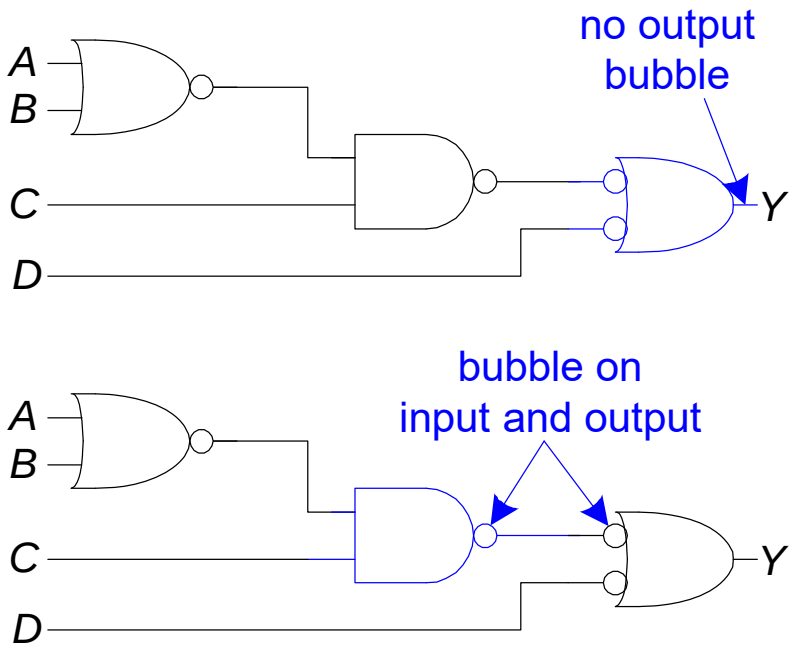
Bubble Pushing Example



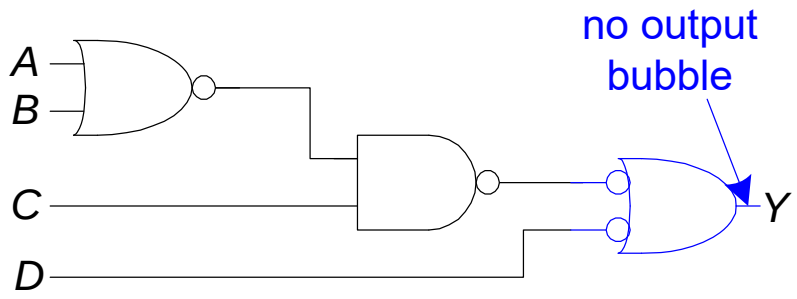
Bubble Pushing Example

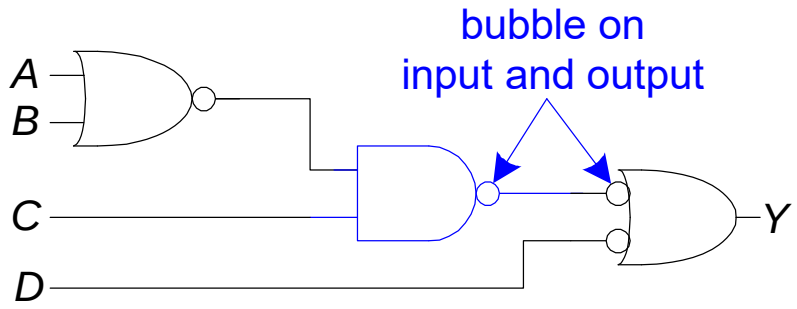


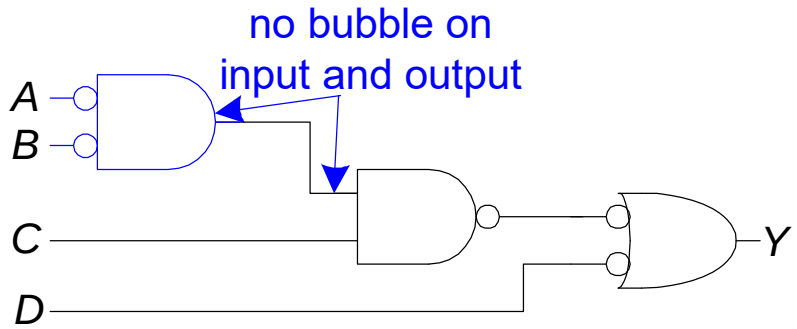
Bubble Pushing Example



Bubble Pushing Example







$Y = \overline{A}\overline{B}C + \overline{D}$

Multiple-Level Optimization

- **Multiple-level circuits - circuits that are not two-level (with or without input and/or output inverters)**
- **Multiple-level circuits can have reduced gate input cost compared to two-level (SOP and POS) circuits**
- **Multiple-level optimization is performed by applying transformations to circuits represented by equations while evaluating cost**

Optimization Example

Optimize the function through transformations, and reduce the gate input cost.

Example: Optimize these multi-level Boolean functions:

$$G = ABC + ABD + E + ACF + ADF \qquad (a)$$

$$G = AB(C + D) + E + AF(C + D) \qquad (b)$$

$$G = (AB + AF)(C + D) + E \qquad (c)$$

$$G = A(B + F)(C + D) + E \qquad (d)$$

(a) gate input cost: 17;
(b) gate input cost: 13;
(c) gate input cost: 12;
(d) gate input cost: 9

Overview

- Introduction
- Binary logic and gates
- Transistors
- Some IC parameters
- Boolean algebra
- Logic functions
- Simplification of logic functions
- Additional Gates and Circuits

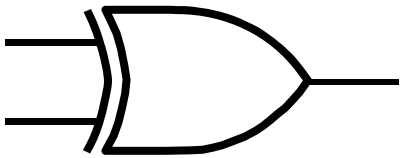
Exclusive OR/ Exclusive NOR

- The eXclusive OR (XOR) function is an important Boolean function used extensively in logic circuits.
- The XOR function may be;
  - implemented directly as an electronic circuit (truly a gate) or
  - implemented by interconnecting other gate types (used as a convenient representation)
- The eXclusive NOR function is the complement of the XOR function
- By our definition, XOR and XNOR gates are complement gates.

Truth Tables for XOR/XNOR

- Operator Rules:

XOR

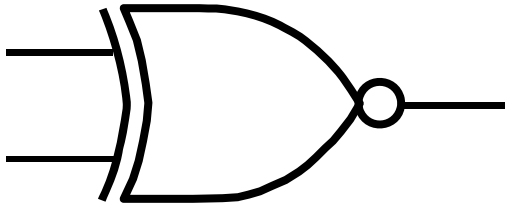


$$X \oplus Y = X\overline{Y} + \overline{X}Y$$

X	Y	$X \oplus Y$
0	0	0
0	1	1
1	0	1
1	1	0

X	Y	$\overline{(X \oplus Y)}$ or $X \equiv Y$
0	0	1
0	1	0
1	0	0
1	1	1

XNOR



- The XOR function means:

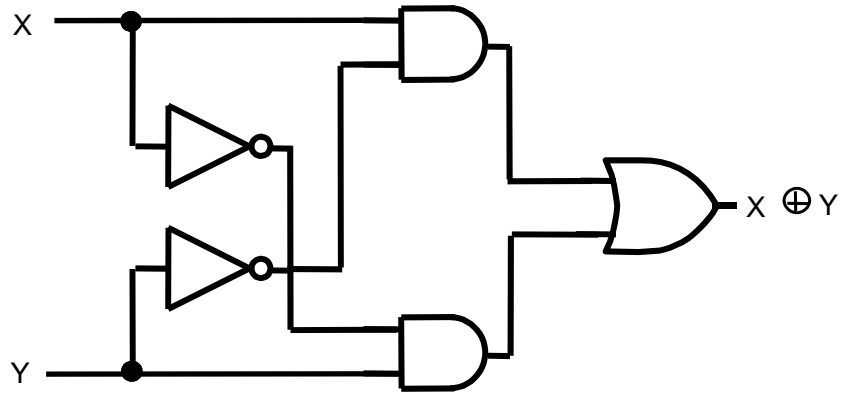
  X OR Y, but NOT BOTH

$$\overline{X \oplus Y} = XY + \overline{X}\,\overline{Y}$$

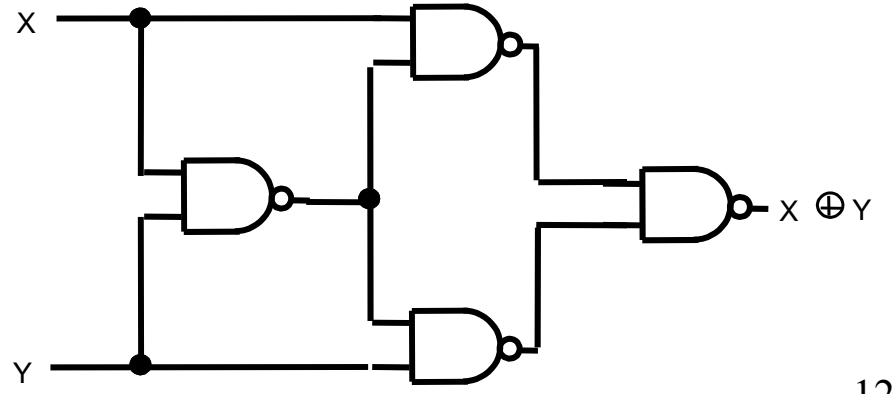
XOR Implementations

- The simple SOP structure:

$$X \oplus Y = X\overline{Y} + \overline{X}Y$$



- A implementation with just NAND is:
  - What's the expression?
  - What's the logic graph?



Application of XOR/XNOR

- XOR / XNOR
  - Adders / subtractors / multipliers
  - Counters / incrementers / decrementers
  - Parity generators / checkers

XOR Identities

$$X \oplus 0 = X \qquad X \oplus 1 = \overline{X}$$

$$X \oplus X = 0 \qquad X \oplus \overline{X} = 1$$

$$X \oplus Y = Y \oplus X$$

$$(X \oplus Y) \oplus Z = X \oplus (Y \oplus Z) = X \oplus Y \oplus Z$$

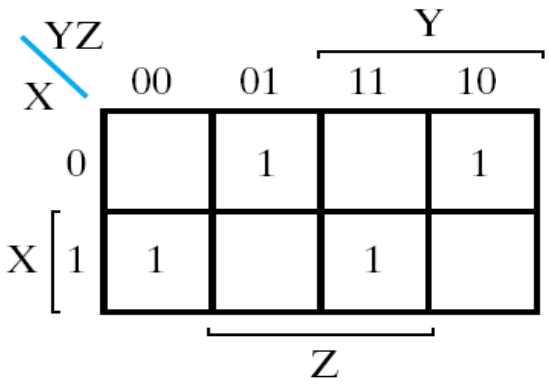
- The complement of the odd function is the even function.
- Symbol $\oplus$ can be replaced by other Boolean equivalence.

$$X \oplus Y \oplus Z = \left(X\overline{Y} + \overline{X}Y\right) \oplus Z$$

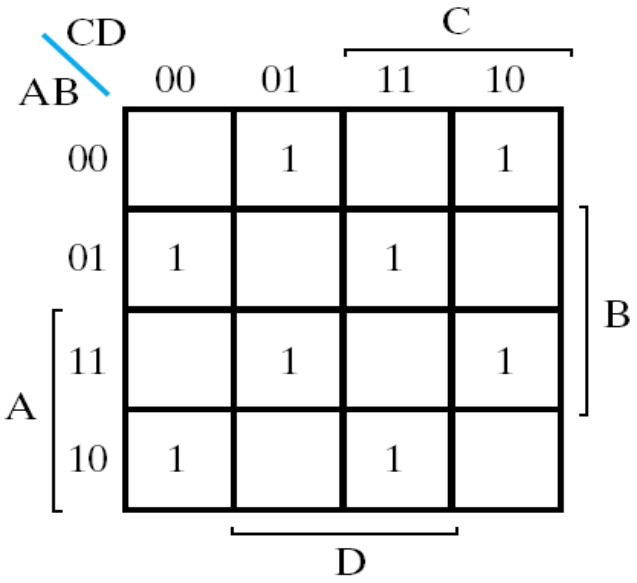
$$= \left(X\overline{Y} + \overline{X}Y\right)\overline{Z} + \left(XY + \overline{X}\,\overline{Y}\right)Z$$

$$= X\overline{Y}\,\overline{Z} + \overline{X}Y\overline{Z} + \overline{X}\,\overline{Y}Z + XYZ$$

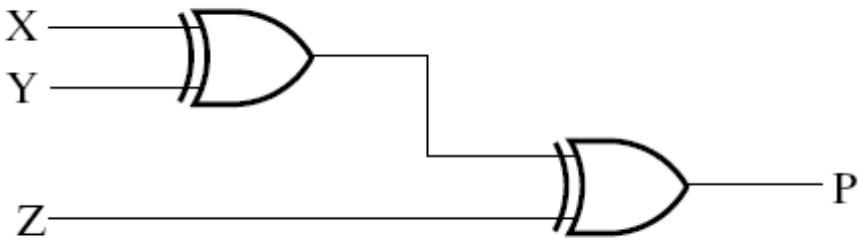
XOR function extended to 3 or more variables.



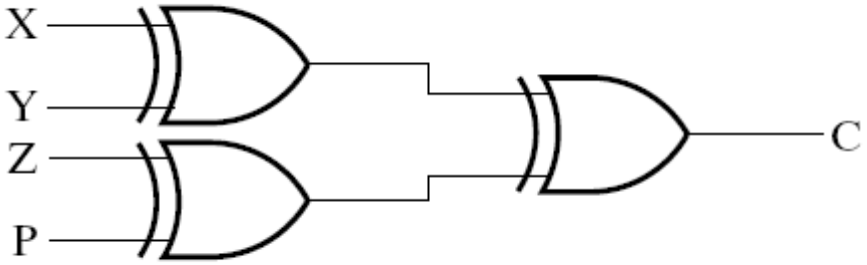
(a) $X \oplus Y \oplus Z$



(b) $A \oplus B \oplus C \oplus D$



(a) $P = X \oplus Y \oplus Z$



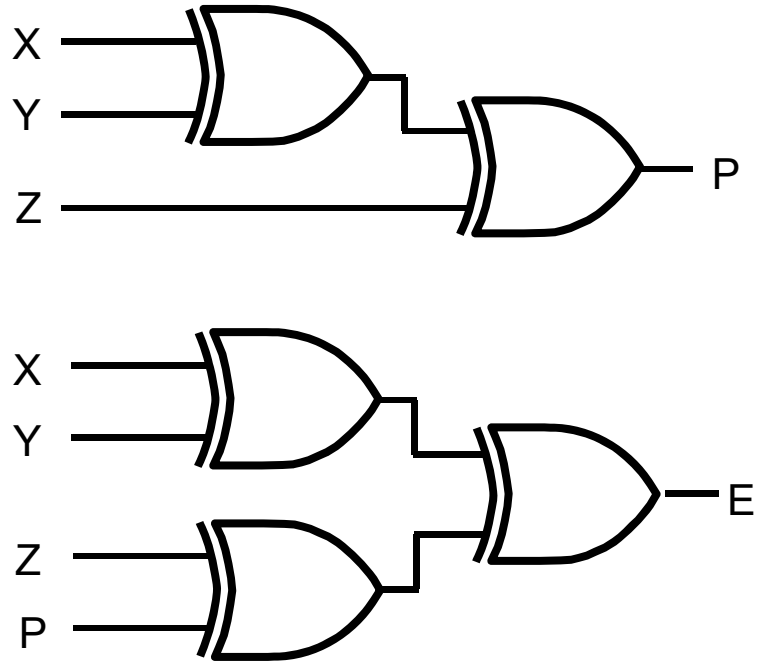
(b) $C = X \oplus Y \oplus Z \oplus P$

Parity Generators and Checkers

- Parity Generators and Checkers
- Parity bit
  - a parity bit added to n-bit code to produce an n + 1 bit code
  - Add odd parity bit to generate code words with even parity
  - Add even parity bit to generate code words with odd parity
- Example: n = 3. Generate even parity code words of length 4 with odd parity generator
- (X,Y,Z) = (0,0,1) gives
  (X,Y,Z,P) = (0,0,1,1) and E = 0.
- Generator: generator parity bit
- Checker: check the change of received data, for example:
  - If Y changes from 0 to 1 between generator and checker, then E = 1 indicates an error.

Parity Generators and Checkers

- Even parity checker in the right circuit
- P
  - parity generator
  - Xyz has odd 1，P=1
  - Xyz has even 1，P=0
- E
  - parity checker
  - E=0，OK
  - E=1，Error



Hi-Impedance Outputs

- Logic gates introduced thus far
  - have 1 and 0 output values
  - cannot have their outputs connected together, and
  - transmit signals on connections in only one direction.
- Imagine:
  - What will happen if the gate outputs are connected together?
- **Cannot Do It!**
- Hi-Z state - Hi-z，Hiz，HiZ

Hi-Impedance Outputs (continued)

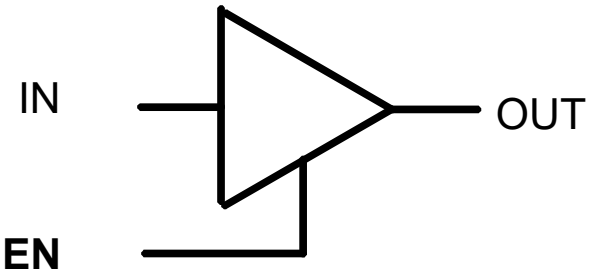
- What is a Hi-Z value?
  - 1 —— high voltage
  - 0 —— low voltage
  - Hi-Z value —— Open circuit，disconnected.
    - It is as if a switch between the internal circuitry and the output has been opened.
- Hi-Z may appear on the output of any gate, but we restrict gates to:
  - a 3-state buffer
  - a transmission gate
- Feature: each of which has one data input and one control input.

The 3-State Buffer

- The 3-State Buffer
  - Symbol and truth table
  - IN, data input,
  - Out， Data Output
  - EN, Control input，Enabled.
- EN = 0

  regardless of the value on IN (denoted by X), the output value is Hi-Z.
- EN = 1　　OUT = IN
- Variations:
  - Data input, IN, can be inverted
  - Control input, EN, can be inverted

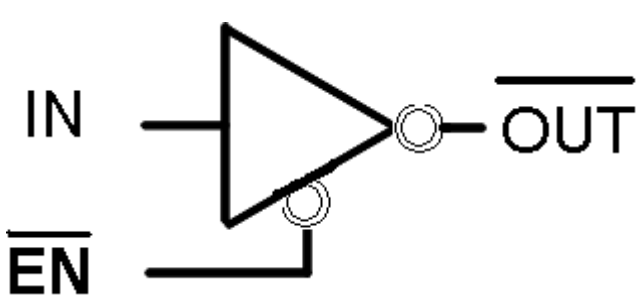
Symbol

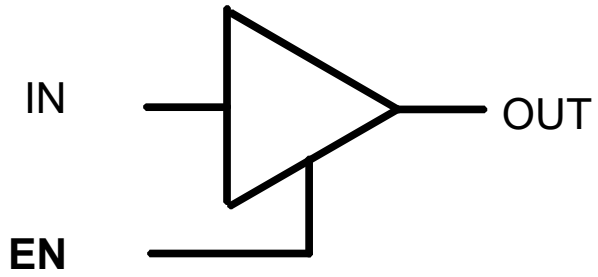


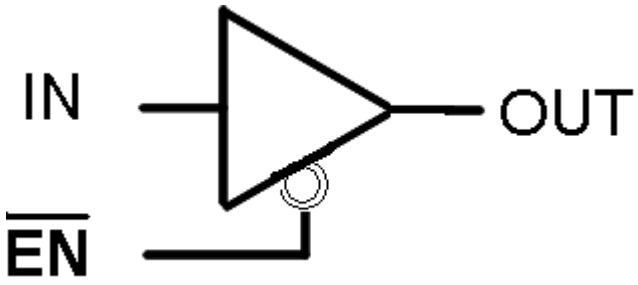
Truth Table

EN	IN	OUT
0	X	Hi-Z
1	0	0
1	1	1

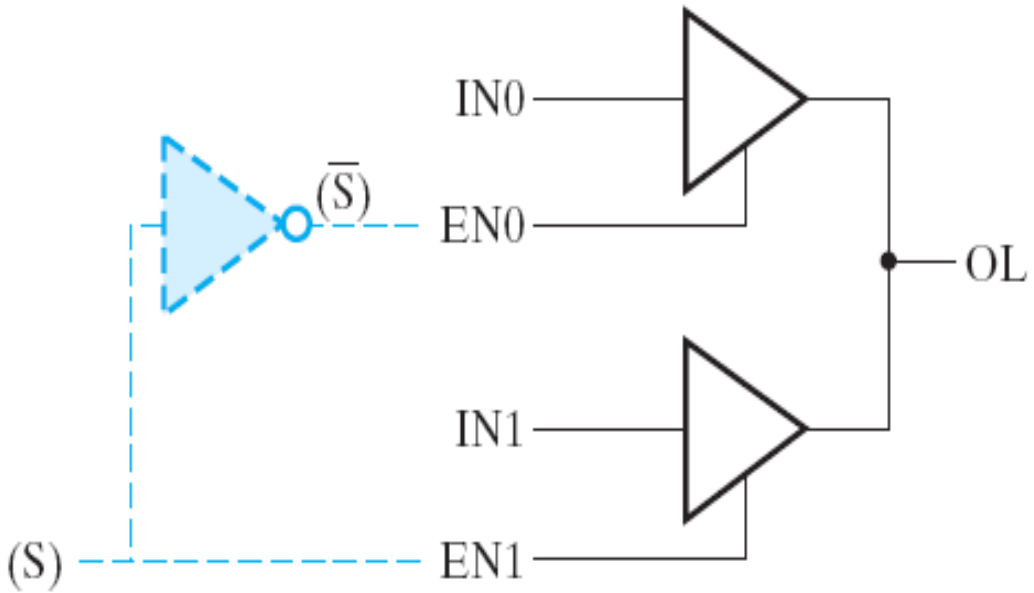
The 3-State Buffer







Multiplexed Line OL



(a) Logic Diagram

EN1	EN0	IN1	IN0	OL
0	0	X	X	Hi-Z
(S) 0	$\overline{(S)}$ 1	X	0	0
0	1	X	1	1
1	0	0	X	0
1	0	1	X	1
1	1	0	0	0
1	1	1	1	1
1	1	0	1	
1	1	1	0	

(b) Truth table

Figure 2-34 Three-state Buffers Forming a Multiplexed Line OL

Transmission Gates

The transmission gate is one of the designs for an electronic switch for connecting and disconnecting two points in a circuit

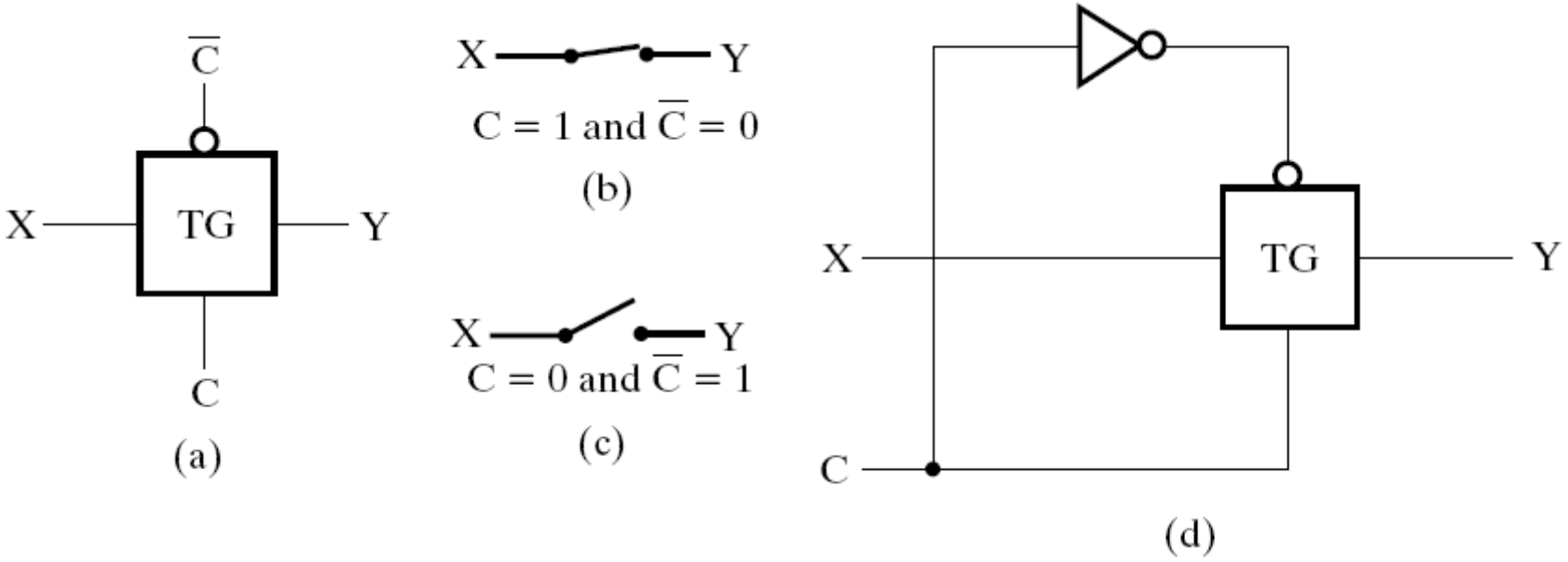
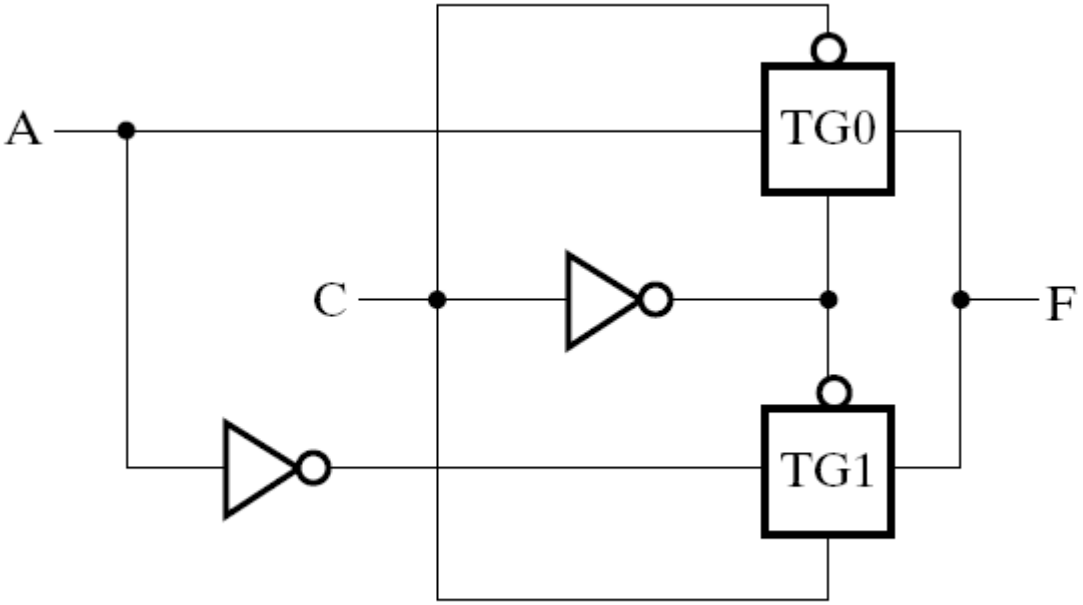


Figure 2-35 Transmission Gates (TG)

Transmission Gates



(a)

A	C	TG1	TG0	F
0	0	No path	Path	0
0	1	Path	No path	1
1	0	No path	Path	1
1	1	Path	No path	0

(b)

Figure 2-36 Transmission Gate Exclusive OR

Summary

- Introduction
- Binary logic and gates
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- Boolean algebra
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- Simplification of logic functions
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